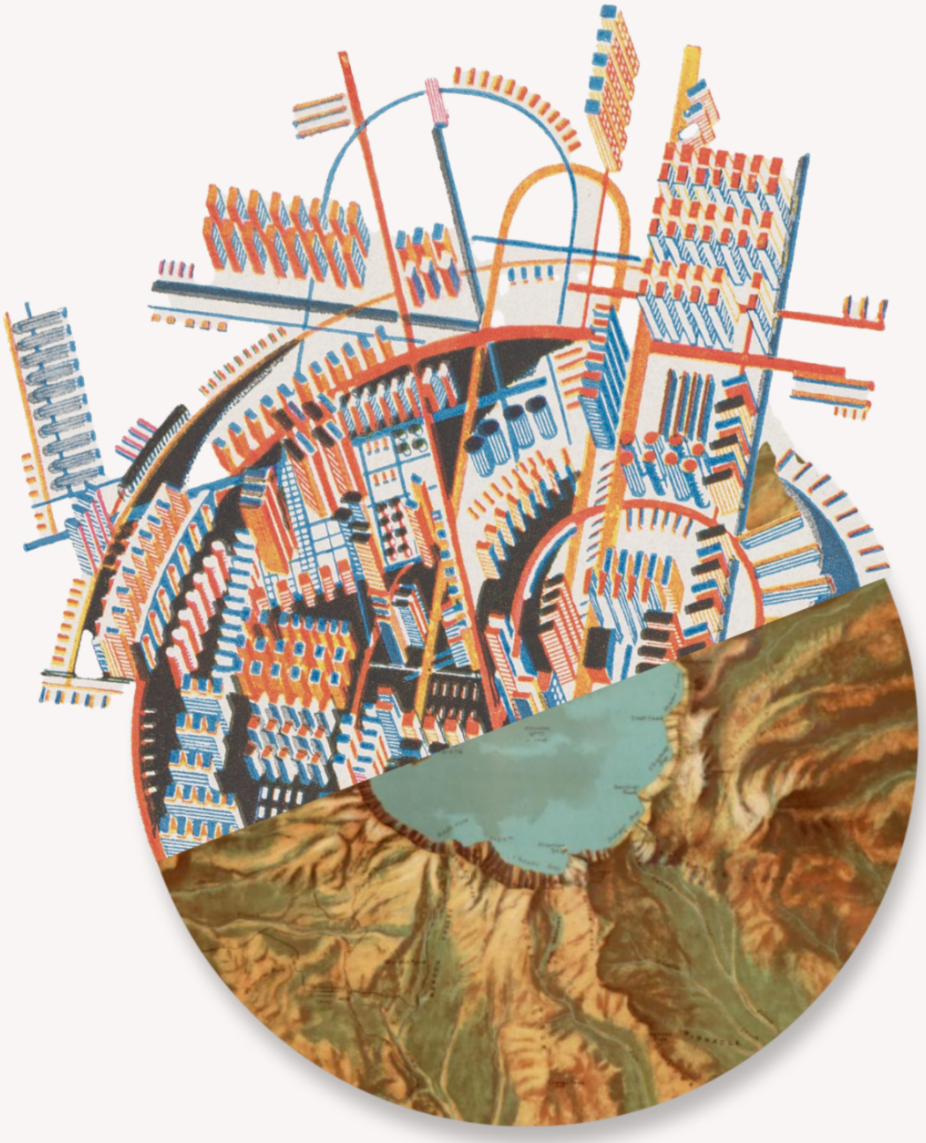

# Emergent Properties

## Decennial Reflections on Collections of the David Rumsey Map Center

# Emergent Properties

# Emergent Properties: Decennial Reflections on Collections of the David Rumsey Map Center

DAVID RUMSEY MAP CENTER, STANFORD UNIVERSITY LIBRARIES, STANFORD, CALIFORNIA

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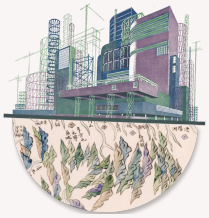
# Introduction

The collections of the David Rumsey Map Center have been shaped by a decade of use and scholarship. As certain materials have resonated with this generation of students, instructors, and researchers, we focus our attention and identify related collections, defining them as distinct clusters within the larger archive. These subsets have in turn defined the David Rumsey Map Center as distinctive, and have set Stanford's collections apart. As a whole, the collection is vast and not bounded by geography, spanning the 15th century to the present. But within that expanse are specific cohesive areas of strength that have added lasting value to campus and beyond.

These clusters emerge through the ways in which we make collections available. Whether through instruction or supporting research, we promote personal inquiry and encourage our users to make their own discoveries. Every opportunity to look at a map is a chance to uncover a new meaning, find a new story, and add to our understanding about the past. We do not lecture, but instead invite users to think critically. As this kind of scholarship evolves, so does the meaning of these objects, and the ways this collection connects with our community.

In 2026 the David Rumsey Map Center celebrated its tenth year—a moment to reflect on how we have added value to scholarship on campus and around the world. Emergent Properties brings together a selection of highlights from deep within the collections of the David Rumsey Map Center: items that have not only proven to be rare or beautiful, but have become the drivers of scholarship, programming, and support areas that have resonated with our users. The forty objects in this exhibition represent these emergent properties. Though only a glimpse into the full breadth of the Center's holdings, they reflect what has mattered most over the past ten years.

# Six Centuries of Mapmaking



The Center's collections offer a broad overview of the history of printed cartography. From early wood-block printed city views, to ornate Dutch maritime atlases, to the evolution of printed maps, to the modern digital era, our holdings encompass a near-infinite combination of maps for teaching and understanding how cartography has been entwined with the human experience and can be read as a reflection of our imagination.

The maps in the Center's collections span six centuries, from the earliest years of the printing press to the present day. They trace the printed map from its earliest stages as the first modern atlas to the colorful pictorial maps of the twentieth century. These maps span the full range of cartographic media, such as manuscript, woodblock, engraving and lithography, each with its own persuasive styles and symbologies.

Instruction is central to the Center's mission, and engaging students with the collection is one of our biggest priorities. The maps that follow invite the questions we explore together in the classroom: How was this map made? Where did its information come from? Who used it, and why? How does it represent land and people? Asking these questions also means seeing maps as products of their time, reflecting the assumptions and biases of their makers alongside their artistry and ingenuity.

G6711 1482 .S8

# 1. Italia di Tolomeo (*Ptolemy's Italy*)

| | |
|---|---|
| **Artist** | Johannes Schnitzer |
| **Place** | Ulm |
| **Year** | 1482 |
| **Dimensions** | 35 × 54 cm (13.8 × 21.3 in.) |
| **Location** | Susan & Benjamin Caughey Map Collection |

The oldest map in our collection is based on the work of Claudius Ptolemy, a 2nd century astronomer and mathematician from Alexandria. His popular work, *Geographia*, listed thousands of known places with their coordinates and explained how to project a round world onto a flat map. None of his original manuscripts have survived and it is uncertain if they even contained maps. Latin translations made in Italy in the early 1400s—some of the earliest examples of printed cartography—contained newly constructed maps which were copied and translated into many languages. This map of Italy, from one of several Ptolemaic atlases printed in Germany starting in the 1470s, was printed from a carved woodblock in black ink, and hand-colored in blue, brown, red, and yellow. Rediscovering Ptolemy helped spark the European Age of Exploration, and his underestimation of the Earth's size is thought to have encouraged Columbus's voyage west to the Indies.

TYPVS ORBIS TERRARVM
SEPTENTRIO.
AMERICA SIVE IN DIA NOVA. An 1492. a phosphoro
Noua Hispania
Tartaria
Mongol
China
India orior.
Arabia
AFRIKA
Nubia
Abissi
Zimbaoe
MAR DEL SVR
OCEANVS AE THIOPICVS.
MAR DI INDIA
Lantchidol mare
Terra del Fuego
TERRA AVSTRALIS NONDVM COGNITA.
MERIDIES.
QVID EI POTEST VIDERI MAGNVM IN REBVS HVMANIS, CVI AETERNITAS OMNIS, TOTIVSQVE MVNDI NOTA SIT MAGNITVDO. CICERO:

# 2. Typus Orbis Terrarum (*Image of the World*)

| | |
|---|---|
| **Artist** | Abraham Ortelius |
| **Place** | Antwerp |
| **Year** | 1570 |
| **Dimensions** | 30 × 42 cm (11.8 × 16.5 in.) |
| **Location** | David Rumsey Map Collection |

This world map combines sophisticated geographic knowledge of some parts of the globe with distortions in others, including nonexistent continents at both North and South poles. It opens the first edition of what is considered the first modern atlas of the world, Abraham Ortelius's *Theatrum Orbis Terrarum* ("Theater of the World"), which contained 53 maps specifically engraved for the volume with a consistent style, format, and lettering, and paired them with text descriptions of the world as Europeans knew it at the time. Its great commercial success set the standard for atlases and launched Ortelius as one of the most prolific of the celebrated Flemish cartographers. Unlike many surviving copies, this map was never hand-colored.

VENETIA.

# 3. Venetia (*Venice*)

| | |
|---|---|
| **Artist** | Georg Braun and Frans Hogenberg |
| **Place** | Cologne |
| **Year** | 1572 |
| **Dimensions** | 34 × 48 cm (13.4 × 18.9 in.) |
| **Location** | David Rumsey Map Collection |

Inspired by the success of Ortelius's *Theatrum*, Georg Braun and Frans Hogenberg created their own compendium of maps and views of the world's cities, titled *Civitates Orbis Terrarum* ("Cities of the World"), which was published in six volumes between 1572 and 1617. Hogenberg had engraved most of *Theatrum's* maps and views, reflecting the close-knit community of mapmakers in northern Europe at the time. This plate from the first volume, devoted mostly to European cities, shows Renaissance Venice with its characteristic canals. It is bordered at the bottom of the page by a numbered list of significant locations, along with a cartouche of contemporary Venetian dress at the center. The composition was based on Jacopo de' Barbari's monumental 1500 *View of Venice*, and for centuries this was the standard way of depicting the city, evidence of how cartographers borrowed freely from those before them.

# 4. Composite: 48 Globe Gores, 2 Polar Calottes, Celestial Globe

| | |
|---|---|
| **Artist** | Vincenzo Coronelli |
| **Place** | Venice |
| **Year** | 1698 |
| **Dimensions** | 26 × 40 cm (10.2 × 15.7 in.) |
| **Location** | David Rumsey Map Collection |

Globes were coveted objects as well as educational tools meant to be browsed, turned, and touched. This composite digitally stitches 48 printed segments into 12 curved gores and two polar end caps, the pieces a globe maker would paste onto a sphere, using papier-mâché techniques, to build this 3.5 feet in diameter celestial globe. The Center holds these printed segments as a bound set, just as a globe maker would have received them. Celestial globes were normally made to show the constellations as if viewed from beyond the stars, looking toward the Earth, reversing how we actually see them. Coronelli instead designed this globe to show the heavens as they appear to an observer on Earth: a “concave” sky. Rather than re-engrave his plates, Coronelli achieved this by countproofing, pulling a second impression from a still wet print to reverse the image. This innovation set this globe apart from his contemporaries.

# 5. Haikou Quantu 海口全圖 (*Complete Map of Sea Ports*)

| | |
|---|---|
| **Artist** | Author unknown |
| **Year** | 1800 |
| **Dimensions** | 30 × 665 cm (11.8 × 261.8 in.) |
| **Location** | David Rumsey Map Collection |

A continuous map of the Chinese coast from Manchuria to Vietnam on a hand-colored scroll reads from right to left and is over twenty feet long, although it would have been read by unrolling just one arm's length at a time. Intended for navigation, the map does not use a uniform scale or grid, and records coastal cities, fortifications, harbors, and reefs along the route. This facsimile section of the scroll shows the area of China and Taiwan, with penciled annotations near the top, where a former owner identified the cities of Zhangzou and Quanzhou in English. Asian cultures developed their own methodologies and practices for depicting geographic space, distinct from those of Europe. Factors like differences in production methods (e.g. printing vs. hand-drawn manuscripts) and collecting practices have resulted in over-representation of European mapmaking traditions in physical archives like this one.

PUBLICADO POR DEBRAY Sucesores — MEXICO

© Cartography Associates, David Rumsey Collection

# 7. Carta Etnografica. II. (*Ethnographic Map II*)

| | |
|---|---|
| **Artist** | Antonio García Cubas |
| **Place** | Mexico |
| **Year** | 1885 |
| **Dimensions** | 55 × 69 cm (21.7 × 27.2 in.) |
| **Location** | David Rumsey Map Collection |

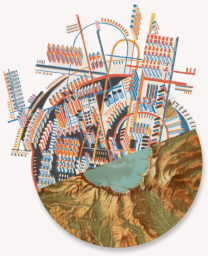
Antonio Garcia Cubas, known to many as the father of Mexican geography, made *Atlas Pintoresco É Historico De Los Estados Unidos Mexicanos* ("Picturesque and Historical Atlas of the United States of Mexico") to convey the identity and character of Mexico. The plates in this atlas celebrate the country's history, industry, landscape, and people, featuring richly detailed color illustrations made with a printing technique called chromolithography. Published as the nation was redefining itself in the decades after the Treaty of Guadalupe Hidalgo ended the Mexican-American war and ceded its northern territories to the United States, the atlas is an homage to Mexico's rich and diverse society. This plate mutes national borders, instead highlighting the territories of the country's distinct ethnic groups, some extending into present-day Arizona and New Mexico, and is framed by vignettes of the peoples and costumes of Mexico's many regions which reflect hierarchical cultural beliefs of the time.

# 8. Americans of Negro Lineage

| | |
|---|---|
| **Artist** | Louise E. Jefferson |
| **Place** | New York |
| **Year** | 1946 |
| **Dimensions** | 73 × 95 cm (28.7 × 37.4 in.) |
| **Location** | David Rumsey Map Collection |

Louise E. Jefferson, born in Washington, D.C. in 1908, was a prominent African American artist who published educational maps about world cultures for classroom use and was the first African American woman to hold a directorship position in the publishing industry in the United States. This map pays homage to the nation's African American heritage, celebrating prominent figures in politics, science, music, and literature alongside statistics showing the many ways African Americans have shaped the country. Jefferson's piece is an example of the genre of pictorial maps, which rose in popularity in the 20th century and used colorful illustrations to educate viewers through a more accessible visual language. Recent scholarship in the history of cartography has brought to light many previously unrecognized contributions to this genre made by women like Jefferson.

# Changing California



The Center's collections trace California's evolution through maps, from the lands of Indigenous peoples, through Spanish and Mexican Alta California, to its incorporation into the United States. These cartographic records chronicle the complex histories and distinctive cultures that persist in California today.

California is deeply integrated into the Center's holdings. Maps of the state where he lived were among David Rumsey's earliest collecting interests, and a large portion of our California material comes from his collection. More recently, Stanford University Libraries and the Bill Lane Center for the American West jointly acquired the collections of the California Historical Society, bringing more than 2,900 historical maps of the state to the Center. Because Stanford's curricula has a distinctly Californian flavor, these materials lend themselves naturally to instruction and engagement and complement the Libraries' many other California collections, from photographs to music to manuscripts.

The maps that follow show California in constant change. An island invented by European imagination, a gold rush that remade the state almost overnight, borders and land ownership redrawn by war and law, and the communities who mapped their own place in the state that has continuously reinvented itself.

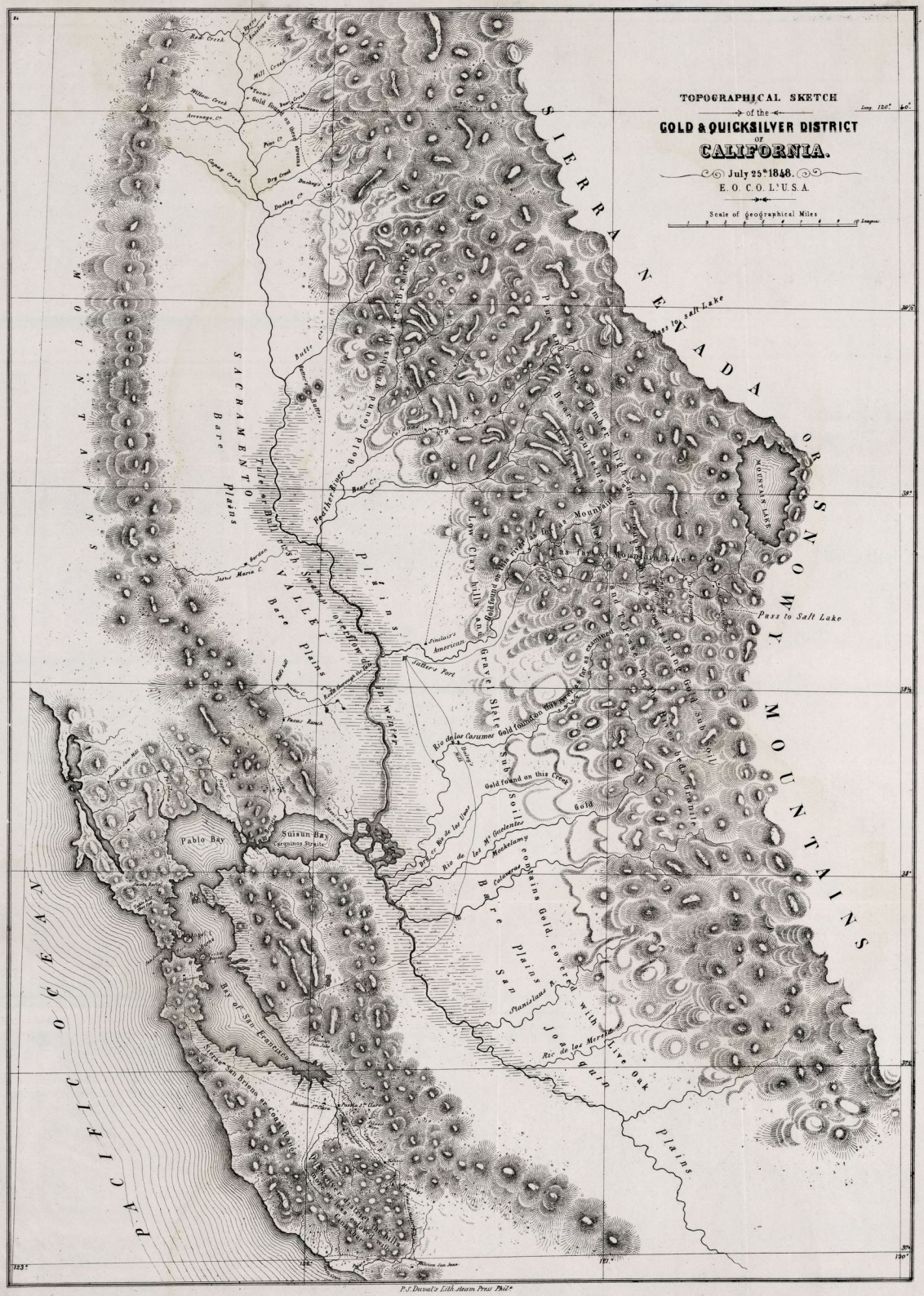
Re-frame

SY109bk9698

# 9. The North part of America conteyning Newfoundland, New England, Virginia, Florida, New Spaine and Nova Francia, with ye riche Iles of Hispaniola, Cuba, Jamaica, and Porto Rieco, on the south, and upon ye west the large and goodly island of California

| | |
|---|---|
| **Artist** | Henry Briggs |
| **Place** | London |
| **Year** | 1625 |
| **Dimensions** | 27 × 34 cm (10.6 × 13.4 in.) |
| **Location** | The Glen McLaughlin Map Collection of California as an Island |

Regarded as the first printed map to show California depicted as an island, this map accompanied an article by the English mathematician Henry Briggs in Samuel Purchas's voyage collection, *Purchas his Pilgrimes.* The idea had roots in a popular Spanish literary work from around 1500 which described an island inhabited solely by women and where gold was the only existing metal. In around 1620 the Dutch captured a document describing California as an island from a Spanish ship. Briggs learned of this and felt confident enough to use it as a source for this map, as the note at lower left explains. This map is part of the Center's Glen McLaughlin Collection, the largest collection of maps depicting California as an Island, with maps dating from 1622 to 1865. These depictions blend fantasy and reality and invite us to ask how this error persisted so long after the true nature of California was known.

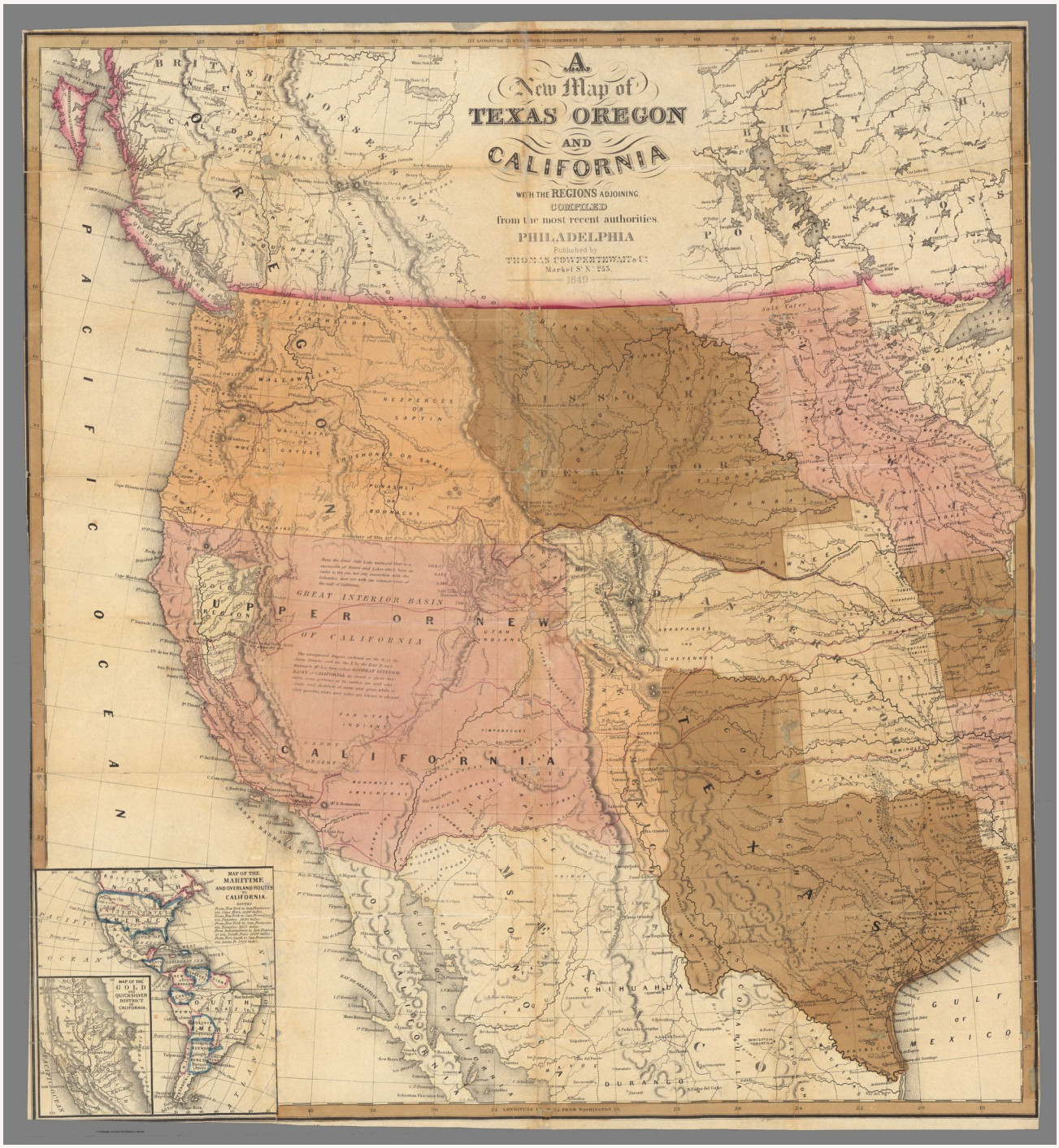



© Cartography Associates, David Rumsey Collection

# 10. Gold and Quicksilver District of California, 1848

| | |
|---|---|
| **Artist** | Edward Otho Cresap Ord |
| **Place** | Washington, D.C. |
| **Year** | 1848 |
| **Dimensions** | 55 × 39 cm (21.7 × 15.4 in.) |
| **Location** | David Rumsey Map Collection |

The Gold Rush transformed California from a military frontier territory into an international cultural and economic center, and maps like this one helped initiate this profound change. Regarded as the first accurate map of gold discoveries in the state, it was published with President Polk's December 5, 1848 message to Congress which, alongside testimonial letters from Californians, helped turn the rumor of gold into a reality. Among that evidence was Thomas O. Larkin's letter written six months after the discovery at Sutter's Mill: "I do not think I am exaggerating in estimating the amount of gold obtained on the rivers... at ten thousands dollars a day for the last few days." On the map, the discoveries appear between Sutter's Fort and Mountain Lake, now Lake Tahoe, at the site labeled "Lower Mine." Prompted by the news, thousands of migrants poured into the state the following year, changing California forever.


A
New Map of
TEXAS OREGON
AND
CALIFORNIA
WITH THE REGIONS ADJOINING.
COMPILED
from the most recent authorities.
PHILADELPHIA
Published by
THOMAS COWPERTHWAIT & Co.
Market St. N. 255.
1849
PACIFIC OCEAN
GREAT INTERIOR BASIN
UPPER CALIFORNIA
NEW MEXICO
CALIFORNIA
TEXAS
MISSOURI TERRITORY
GULF of MEXICO
MAP OF THE MARITIME AND OVERLAND ROUTES TO CALIFORNIA.
MAP OF THE GOLD AND QUICKSILVER DISTRICT OF CALIFORNIA.

# 11. A new map of Texas, Oregon and California with the regions adjoining: Compiled from the most recent authorities

| | |
|---|---|
| **Artist** | Samuel Augustus Mitchell |
| **Place** | Philadelphia |
| **Year** | 1849 |
| **Dimensions** | 57 × 52 cm (22.4 × 20.5 in.) |
| **Location** | David Rumsey Map Collection |

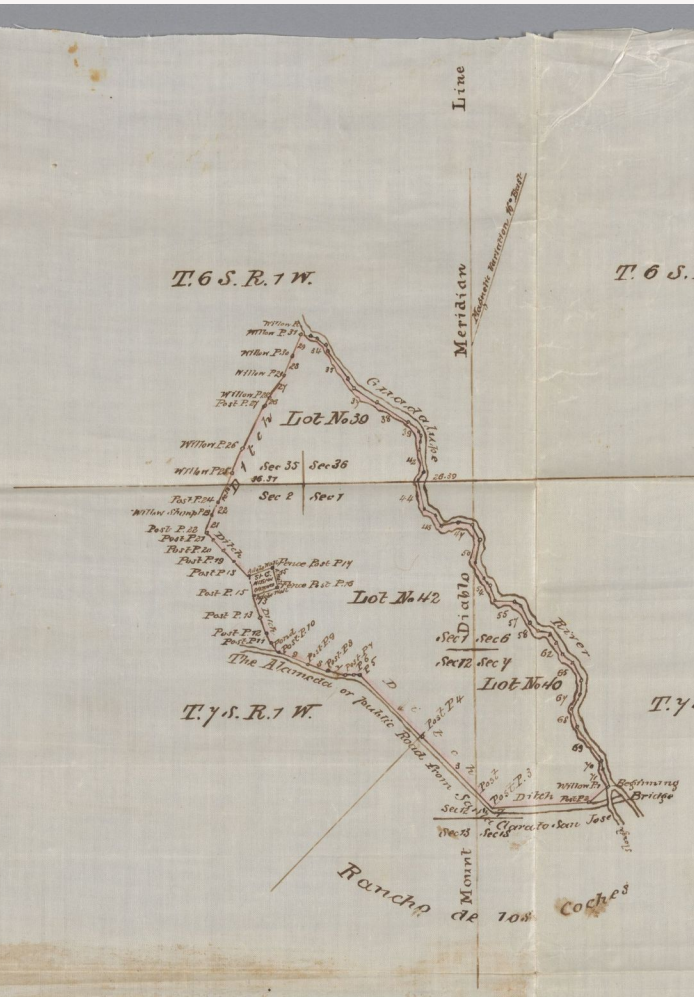
The year after the Treaty of Guadalupe Hidalgo ended the Mexican-American War and transferred the Southwest to the United States, this pocket map contextualized the newest land acquisitions in the shifting borders of the vast American West. Secluded on the coast, Upper California already hints at its future stature. The “Gold Regions,” highlighted in faded yellow ink, stand out as the territory’s major feature. One of its two insets charts the maritime and overland routes from the populated eastern seaboard, journeys that could run as long as the 17,000 mile passage around Cape Horn at the southern tip of South America. The other details the gold and quicksilver district itself. Folded into leather covers, it was made to travel west in a forty-niner’s pocket.

Заливъ Нидейцов
Широта 40° 45'
М. Южное Долгота 124° 05'
Склоненіе 17° 0
КАРТА
Сѣверо западнаго берега
АМЕРИКИ
отъ Широты 39° до 44°
исправленная астрономическими наблюденіями
Ново Архангельскъ
1848
XIII
М. Горнаго
М. Григорія
М. Телло
М. Орфордъ
М. Северный
М. Южный
М. Мендодино
М. Вискаино
Мысъ Григорій на N.O 41° въ 5 Миляхъ
Мысъ Орфордъ на N.O 5° въ 10 Миляхъ
Мысъ Мендодино Южн. Предълъ на N.O 18° въ 25 Миляхъ
Ск. Кол. 18° О
Ск. Кол. 17° О
Долгота Западная отъ Гринича

## 12. Карта сиеверо западного берега Америки от ширины 39° до 44° исправленнаиа Колониальными мореходцами (*Map of the northwest coast of America from latitude 39° to 44° corrected by Colonial mariners*)

| | |
|---|---|
| **Artist** | Mikhail Dmitrievich Tebenkov |
| **Place** | Sitka |
| **Year** | 1848 |
| **Dimensions** | 54 × 43 cm (21.3 × 16.9 in.) |
| **Location** | David Rumsey Map Collection |

California's history is usually told as being a Spanish to Mexican to American transition, but this chart is evidence of a third imperial player. In the first half of the 19th century Russian expansion pushed south from Alaska along the coast, mainly drawn by the sea otter fur trade. Fort Ross, just south of the area shown, marked the southernmost Russian settlement in California. Prepared in Sitka, the capital of Russian America, this chart covers the coast from southern Oregon to near Fort Bragg. The three headland views on the right were meant for recognizing landmarks from the water. With an inset of Humboldt Bay, its penciled navigation calculations suggest it might have actually been used aboard a Russian ship. Its title notes that it was "corrected by Colonial mariners," incorporating knowledge gathered on the spot. With imperial priorities in Europe and otter populations collapsing from overhunting, a stronger Russian California never came to be.




# PLAT

OF THE RANCHO

# EL POTRERO DE SANTA CLARA

Finally confirmed to

Robert F. Stockton

surveyed under instructions from the

U. S. Surveyor General

by

Wm. J. Lewis. Dep. Surv.r

December 1860

Containing 1939 ⁷⁄₁₀₀ Acres

Scale 40 chains to 1 inch

## Boundaries

| No. | Course | Distance | No. | Course | Distance | No. | Course | Distance |
|---|---|---|---|---|---|---|---|---|
| 1 | S 30 ¾ W | 10.50 | 25 | W 2 ¼ E | 23.00 | 49 | S 33 ¼ E | 9.00 |
| 2 | S 86 ¾ W | 45.63 | 26 | N 30 ¼ E | 4.60 | 50 | S 45 ¾ W | 6.00 |
| 3 | N 44 ¾ W | 43.00 | 27 | N 35 ¼ E | 12.43 | 51 | S 25 ½ W | 4.50 |
| 4 | N 44 W | 43.16 | 28 | N 11 ½ E | 10.50 | 52 | S 24 ½ E | 9.00 |
| 5 | S 84 ¼ E | 3.00 | 29 | N 30 ¼ E | 14.87 | 53 | S ½ E | 5.00 |
| 6 | N 59 ¼ W | 3.50 | 30 | N 54 ½ E | 2.00 | 54 | S 13 E | 8.00 |
| 7 | N 24 ½ W | 6.80 | 31 | N 37 ½ E | 3.50 | 55 | S 83 ½ E | 9.20 |
| 8 | N ½ W | 10.00 | 32 | N 42 E | 3.00 | 56 | S 65 E | 3.50 |
| 9 | N ¾ W | 19.19 | 33 | N 5 ¼ E | 6.90 | 57 | S 24 E | 9.00 |
| 10 | N 18 W | 3.00 | 34 | N 84 ¼ E | 3.50 | 58 | S 36 ½ E | 6.00 |
| 11 | N 2 ¾ W | 5.00 | 35 | S 3 ½ E | 16.30 | 59 | S 85 ½ E | 5.00 |
| 12 | N 2 ½ W | 8.00 | 36 | S 26 ½ E | 4.60 | 60 | S 61 E | 3.00 |
| 13 | N 16 ½ W | 11.19 | 37 | S 5 ¼ E | 13.00 | 61 | S 39 ¼ E | 3.00 |
| 14 | N ¼ E | 12.00 | 38 | S 5 ¼ E | 18.00 | 62 | S 27 E | 8.00 |
| 15 | N 11 W | 0.27 | 39 | S 46 ¼ E | 8.00 | 63 | S 52 ¼ E | 5.00 |
| 16 | S W | 17.36 | 40 | S 4 ¼ E | 2.20 | 64 | S 35 ¼ E | 4.00 |
| 17 | N 10 ½ W | 8.00 | 41 | S 7 ¼ W | 44.55 | 65 | S 71 ½ E | 7.00 |
| 18 | N ½ W | 8.00 | 42 | N 25 ¼ E | 12.80 | 66 | S 01 ¾ W | 4.95 |
| 19 | N W | 7.50 | 43 | N 37 ¼ W | 5.00 | 67 | S 27 ¼ W | 9.90 |
| 20 | N 10 W | 5.00 | 44 | S 7 ¼ E | 15.50 | 68 | S 74 ¼ E | 7.03 |
| 21 | N 18 ½ E | 8.84 | 45 | S 63 E | 2 ¼ | 69 | S 36 ¼ E | 21.00 |
| 22 | N 2 ½ E | 5.45 | 46 | S 2 ¾ W | ⅞ | 70 | S 41 ½ E | 2.00 |
| 23 | N 26 ¼ E | 19.00 | 47 | N 8 ½ E | ⅞ | 71 | S 27 ¼ E | 6.00 |
| 24 | N 2 ¼ E | 11.50 | 48 | S ¾ E | ⅞ | 72 | S 46 ¼ W | 3.92 |
| | | 320.16 | | | 208.05 | | | 150.43 |
| | | | | | | | | 208.05 |
| | | | | | | | | 320.86 |

A aggregate of exterior Boundaries 686.36 = 8 Miles 46 chains 36 ⁷⁄₁₀ links

Note: Notice of the approval of this plat and its retention in office for four weeks subject to inspection, was published from 3rd of February to 22nd of March in pursuance of the act of Congress of June 14. 1860

J. W. Mandeville
U. S. Surv.r Gen.l Calif.

The fieldnotes of the Rancho "El Potrero de Santa Clara" and from which this Plat has been made, have been examined and approved and are on file in this office

U. S. Surveyor Generals office
San Francisco, California
January 20th 1861

J. W. Mandeville
U. S. Surv.r Gen.l Calif.

A full true and correct copy of the original Plat on file in this office
U. S. Surveyor Generals office
San Francisco, California
June 29th 1861

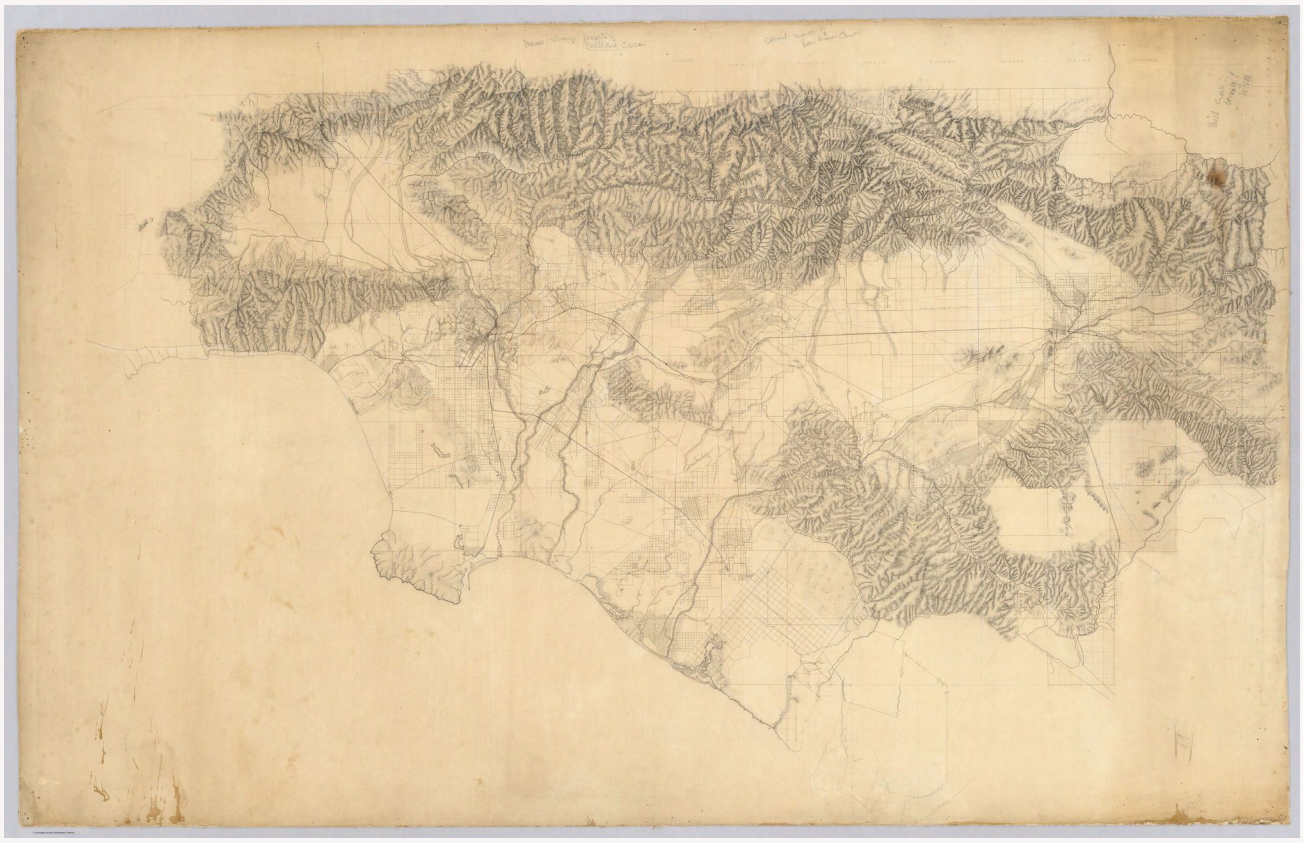
E. F. Beale
U. S. Surv.r Gen.l Calif.

L. S.

# 13. Plat of the Rancho el Potrero de Santa Clara

| | |
|---|---|
| **Artist** | James W. Mandeville |
| **Year** | 1861 |
| **Dimensions** | 48 × 37 cm (18.9 × 14.6 in.) |

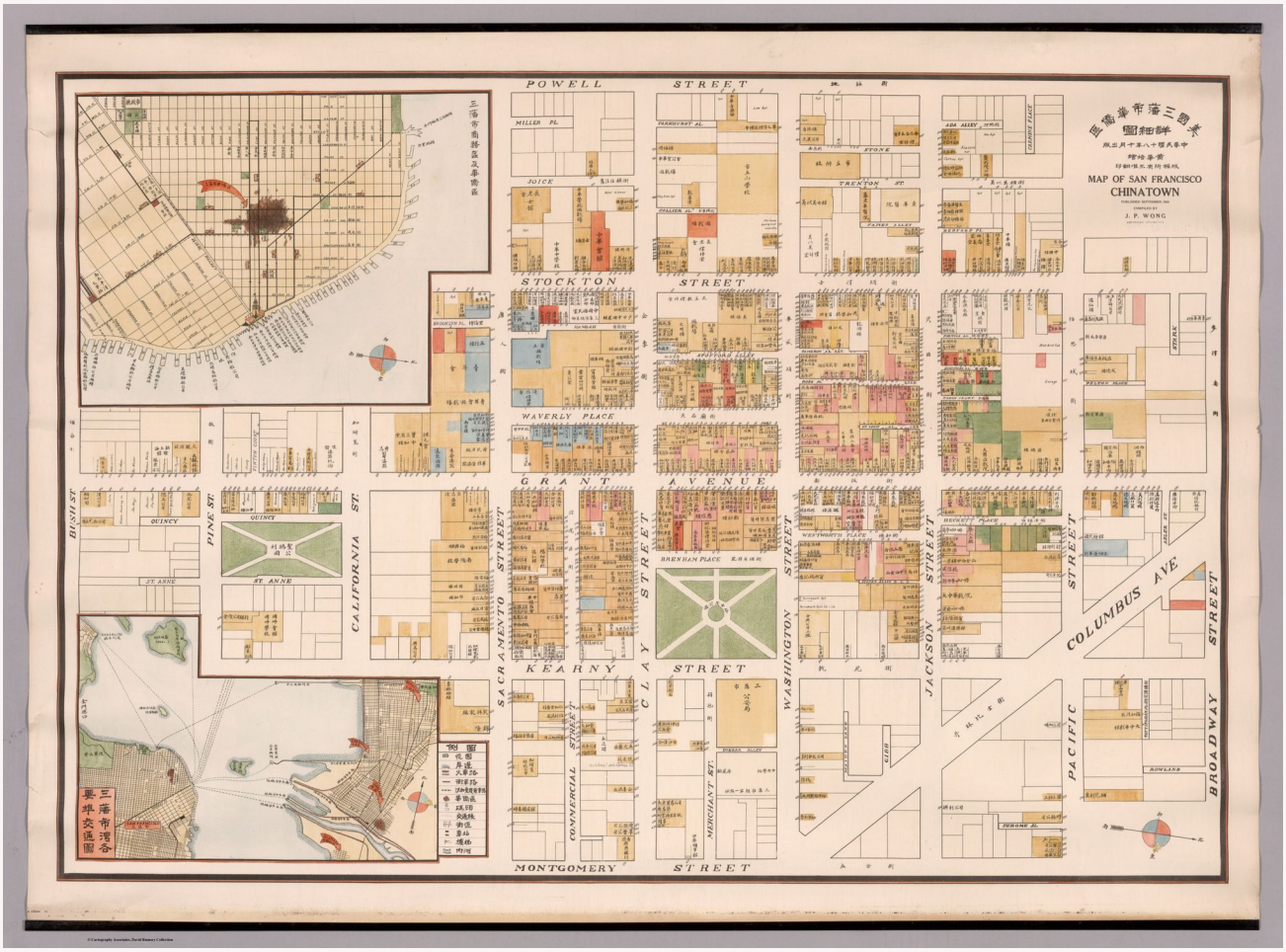
The Treaty of Guadalupe Hidalgo promised that legitimate Mexican land claims would be honored in the new state of California, but in practice landowners had to prove their ownership in court. Under Mexican law, a petition only needed a diseño: a rough sketch describing boundaries by landmarks. American courts demanded more, and the costly surveys, decades-long proceedings, and disputes that followed cost many Californio families—Hispanic Californians who had settled in California under Mexican and Spanish rule—their land and livelihoods. This manuscript plat shows the process's end stage, with precisely measured boundaries and distances, and was made once the professional surveying industry had established itself in the state. This map confirms ownership of 1,939 acres along the Guadalupe River near San Jose, granted to James Alexander Forbes by Governor Micheltorena in 1844 and later acquired in 1867 by Robert F. Stockton, a central figure in the early American occupation of California.



# 14. Los Angeles - San Bernardino Basin, circa 1880

| | |
|---|---|
| **Artist** | William Hammond Hall |
| **Year** | [1880] |
| **Dimensions** | 93 × 147 cm (36.6 × 57.9 in.) |
| **Location** | David Rumsey Map Collection |

The dry Southern California climate limited the growth of major cities until water sources could be developed. William Hammond Hall, designer of Golden Gate Park in San Francisco, and California's first State Engineer, oversaw early state water projects from flood control to irrigation, and this hand-drawn basin map was his working reference copy and likely the base for the maps in his 1888 report on irrigation in California. Penciled notes indicate a working document, such as the memo at the top: "change mouth of Millard Cañon." It captures the region during a transformative time, when railroads like the Atchison, Topeka and Santa Fe Railway crossed the basin, bringing newcomers drawn to the area's climate and opportunities. The scattered street grids of separate towns are like islands in the open, dramatically different from the continuous urban landscape we know today.




© Cartography Associates, David Rumsey Collection

# 15. Mei guo san fan shi hua qiao qu: xiang xi tu (*Map of San Francisco Chinatown*)

| | |
|---|---|
| **Artist** | J.P. Wong |
| **Place** | San Francisco |
| **Year** | 1929 |
| **Dimensions** | 47 × 67 cm (18.5 × 26.4 in.) |
| **Location** | David Rumsey Map Collection |

This is the first map of San Francisco made by Chinese Americans for the Chinese community. Chinese immigrants were among the first to arrive in Gold Rush California, and many later built the transcontinental railroad. However they were scapegoated as taking jobs from white laborers, and anti-Chinese sentiment brought violence that escalated to the creation of the Chinese Exclusion Act barring new Chinese laborers from entering the United States in 1882. Exclusion spurred stronger community institutions like the Chinese Consolidated Benevolent Association, for whom this map was made. Written almost entirely in Chinese, it closely resembles a map the City of San Francisco made in 1885 to investigate Chinatown's living conditions. Four decades later, that outsider's survey is reborn as an insider's portrait of a cohesive neighborhood, with an inset map showing ferry connections to Oakland's Chinatown across the Bay.

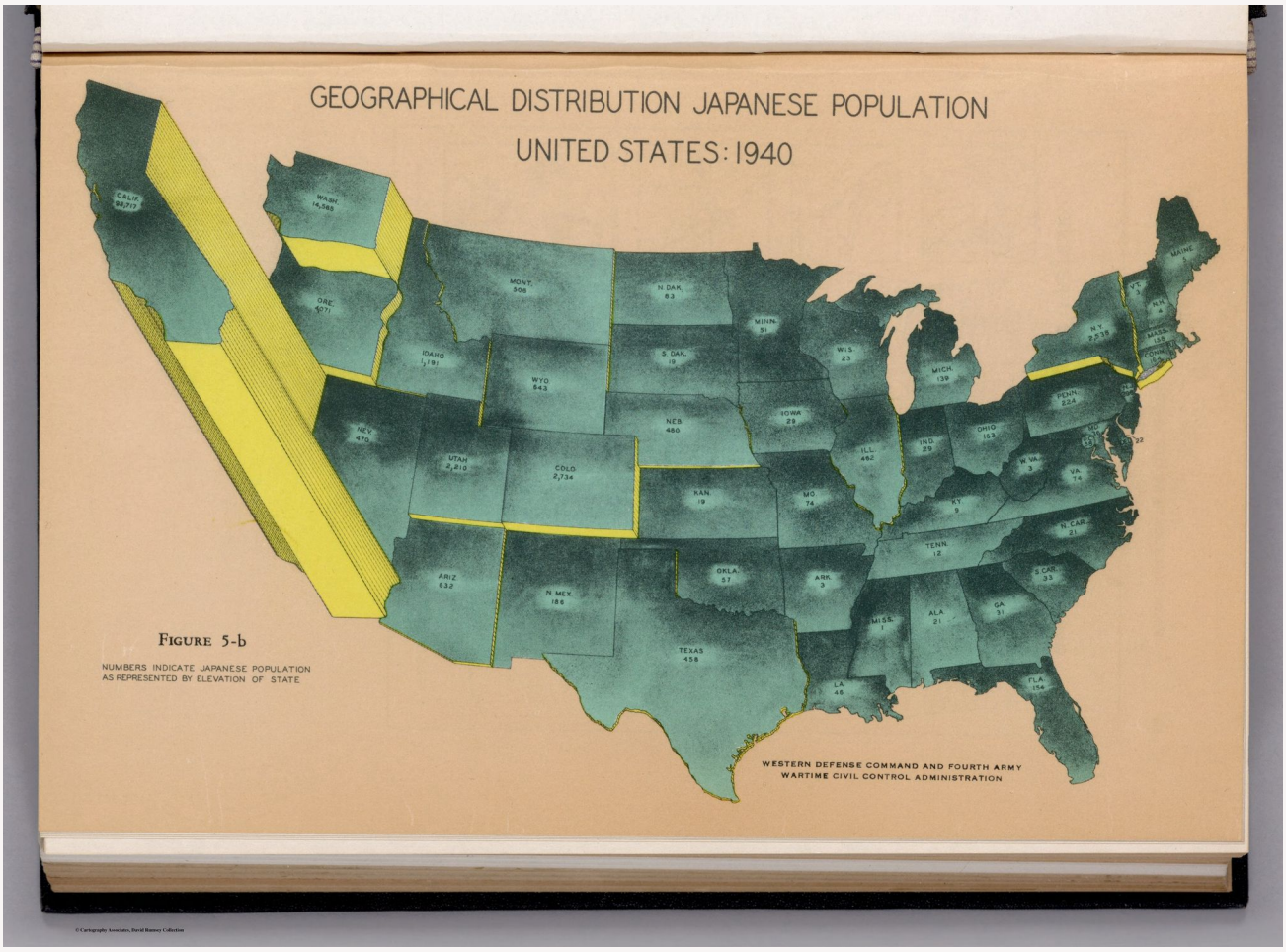



FIGURE 5-b

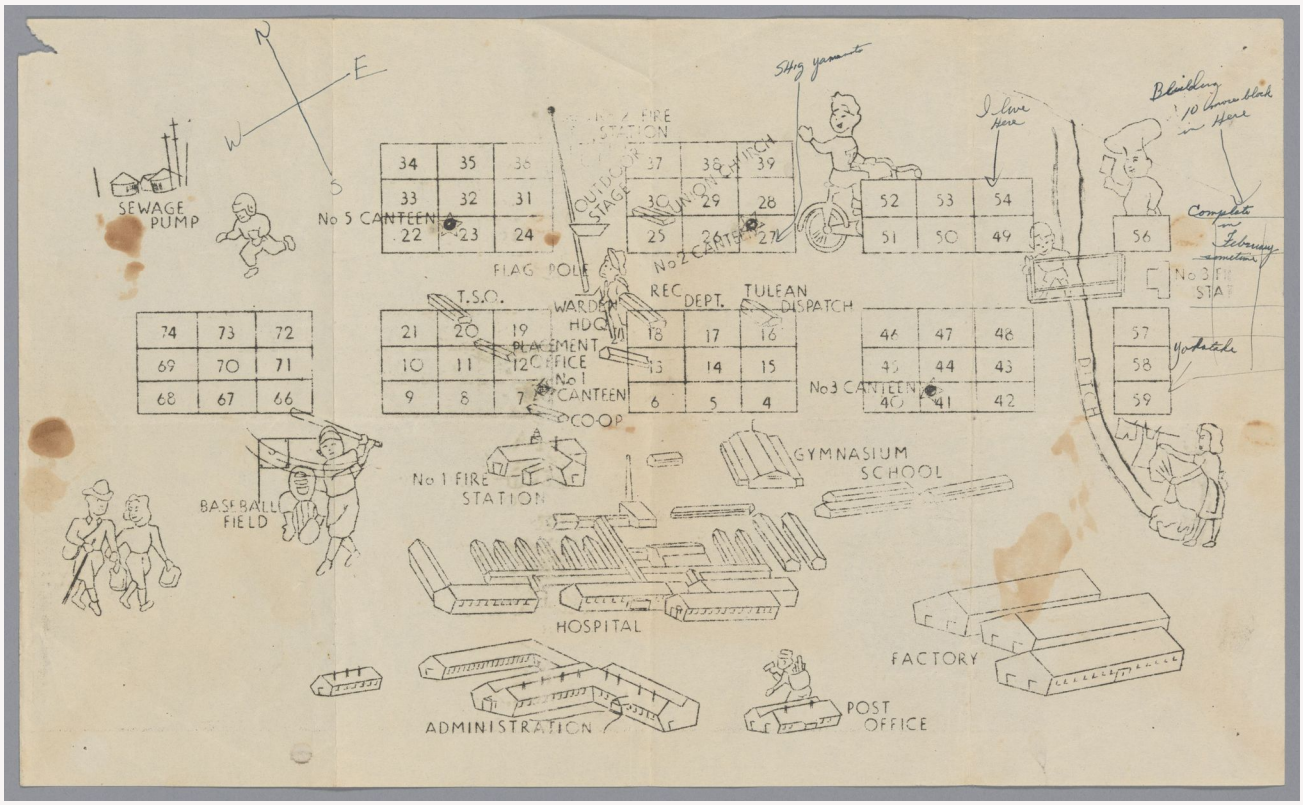
NUMBERS INDICATE JAPANESE POPULATION AS REPRESENTED BY ELEVATION OF STATE

© Cartography Associates, David Rumsey Collection

# 16. Geographical Distribution Japanese Population. United States: 1940

| | |
|---|---|
| **Artist** | United States Office of War Information |
| **Place** | Washington, D.C. |
| **Year** | 1943 |
| **Dimensions** | 15 × 23 cm (5.9 × 9.1 in.) |
| **Location** | David Rumsey Map Collection |

Driven by hysteria after the bombing of Pearl Harbor, the United States adopted policies that forced Japanese Americans, regardless of whether they were U.S. citizens, into relocation centers. This figure comes from the War Department's final report on the removal, *Japanese evacuation from the West Coast, 1942 : Final report,* a volume of 43 maps and numerous charts assembled to justify the incarceration as "military necessity." Using 1940 census data, it extrudes each state to indicate the size of its Japanese American population, making California tower over the rest of the country and fueling the perception that the state needed to be brought under control. It is a reminder that maps do not only record events, but can also be designed to argue for them. In the 1980s it was found that this report had been doctored before reaching the courts, which helped retroactively overturn wartime cases against Japanese Americans for violating curfew and exclusion orders.


N
E
W
S
SEWAGE PUMP
No 5 CANTEEN
34 35 36
33 32 31
22 23 24
FLAG POLE
OUTDOOR STAGE
37 38 39
30 29 28
25 26 27
No 2 CANTEEN
52 53 54
51 50 49
56
I live here
Sling yarn
Complete in February sometime
No 3 FIRE STA
57
58
59
74 73 72
69 70 71
68 67 66
T.S.O.
WARDEN HDQ
REC DEPT.
TULEAN DISPATCH
21 20 19
10 11 12
9 8 7
PLACEMENT OFFICE
No 1 CANTEEN
COOP
18 17 16
13 14 15
6 5 4
46 47 48
45 44 43
40 41 42
No 3 CANTEEN
BASEBALL FIELD
No 1 FIRE STATION
GYMNASIUM SCHOOL
HOSPITAL
FACTORY
ADMINISTRATION
POST OFFICE

# 17. [Map of Tule Lake Relocation Center]

| | |
|---|---|
| **Artist** | Tulean Dispatch Daily |
| **Place** | Newell, California |
| **Year** | 1943 |
| **Dimensions** | 21.6 × 36 cm (8.5 × 14.2 in.) |

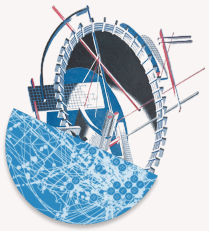
The War Department's report mapped Japanese Americans as a problem to be managed, but this map was made by the incarcerated themselves. At Tule Lake, in the far northeastern corner of California near the border with Oregon, more than 7,000 people were incarcerated. This simple map, printed on legal-sized paper with the few resources at hand, was issued by the Tulean Dispatch, a newspaper published to keep residents of the camp informed of events, announcements, and news from other camps. Its numbered blocks map where people lived, while its drawings show how they made the camp as much like home as they could: a couple holding hands, baseball players, a bicyclist, the school, hospital, and post office. A former resident annotated this copy by hand, marking friends' barracks and, beside Block 54, the words "I live here."

# 18. California

| | |
|---|---|
| **Artist** | David Schiller |
| **Place** | San Jose |
| **Year** | 1968 |
| **Dimensions** | 95 × 71 cm (37.4 × 28 in.) |
| **Location** | David Rumsey Map Collection |

Three and a half centuries after Briggs, California again floats free from the continent, this time by choice. David Schiller, one of the kings of 1960s rock art who created album covers and posters for psychedelic rock bands, drew the state through the lens of its own counter culture, depicting it as a sort of island utopia detached from the rest of the world. His exaggerated landscape celebrates California's striking natural diversity, from the jagged Sierra Nevada to the coastal redwood forests, and is packed with the era's famous venues and landmarks. Originally created for Jefferson Airplane, the map shows the band flying over the mountains and scattering flowers in its path.

# Mapping Africa: Colonial Views to Local Narratives



Maps of Africa, especially concentrated on South Africa, are a cornerstone of the Center's holdings, containing thousands of maps and atlases. Nearly all of them were made by Europeans, for European eyes. They map the continent as a coastline to be charted, a mystery to be discovered, and a prize to be claimed. But looking deeply, local places, communities, knowledge, and histories can surface even in maps that were meant to serve an empire.

This collection has grown from notable acquisitions, among them the collections of Dr. Oscar I. Norwich, the Johannesburg collector whose maps of Africa and its islands anchor the holdings, and of Caroline Batchelor. Academic advocates such as Professor Grant Parker have extended its reach by facilitating digital additions from partners including the University of Cape Town. This collection holds few maps of Africa made from African perspectives, a gap in the cartographic record itself, and a priority for future acquisitions into the Center's holdings.

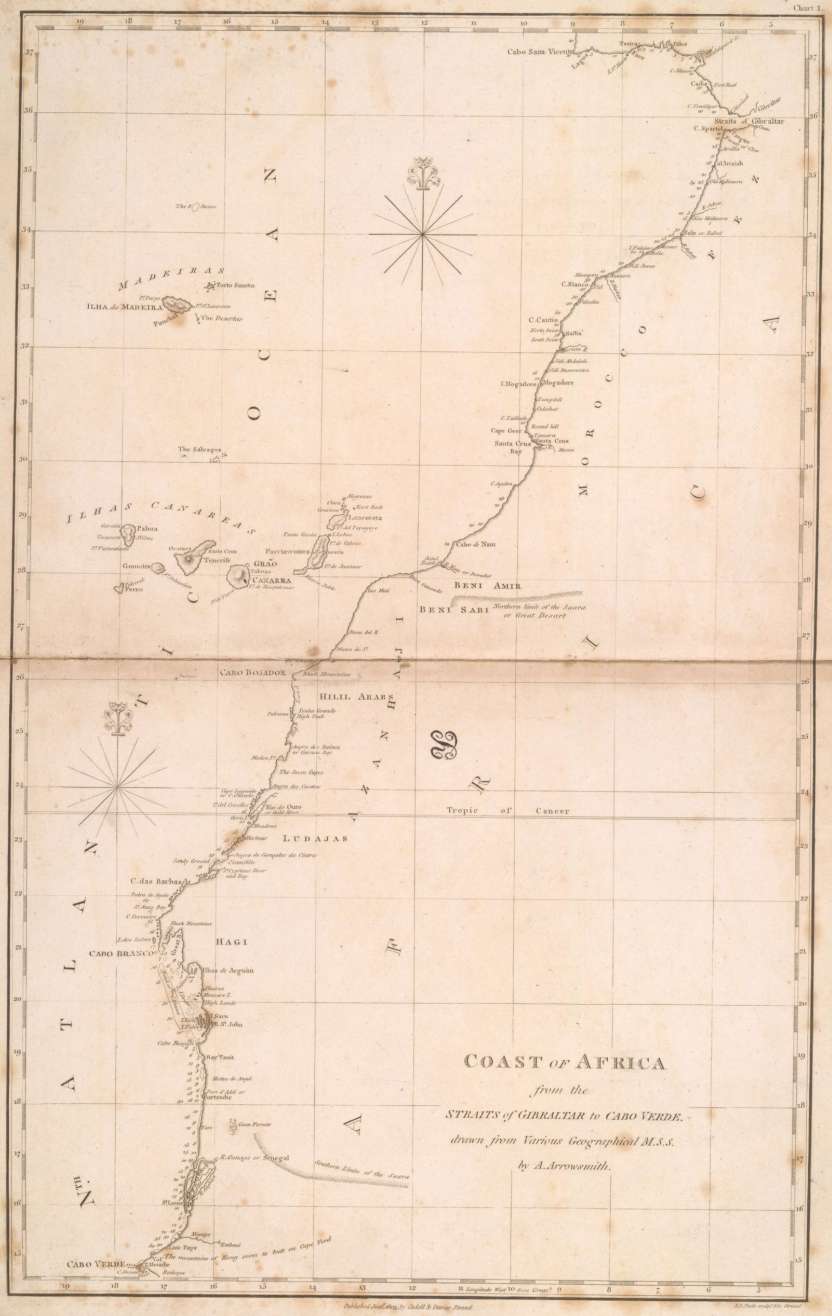
The maps that follow move from the first atlases of the continent through exploration and empire, and close with a contemporary artist's vision of an Africa never colonized at all.

ASIAE PARS.
REGNI PERSIDE PARS
OCEANVS ATLANTICVS
MARE MEDITERRANEVM
MARE DI MELEGNIN
MARE ROSSO
AYMAN olim Arabia felix.
TROPICVS CANCRI
AEQVINOCTIALIS CIRCVLVS.
MAR DEL NORT.
NIGRITARVM REGIO.
NVBIA
BILEDVLGERID.
ZANGVAR
BIAFAR
MEDRA
BENIN.
GAGO.
MELLI
TOMBVTO
GVBER.
GOYAME.
DAMVT.
DAFILA
BARNAGASSO.
BELLO
MAGADAZO
OLABI
BELLEGVAN
SINVS BARBARICVS, qui et Arabicus vulgo mare Rubrum
RVBRVM MARE
OCEANVS AETHIOPICVS
PARS BRASILIAE
AFRICAE TABVLA NOVA.
TROPICVS CAPRICORNI
Canarie insule olim Fortunate.
Insule Cap. Viridis olim Gorgades siue Hesperides
Rio de la plata, el rio argentino fiume
Hic longe latoq. imperitat magnus princeps Presbyter Ioannes totius Africæ
Hæc insula ab incolis Madagascar ab Hispanis S. Laurentij, olim Mirabilis Ptol. Cerne sortè Pluuiorum.

# 19. Africae tabula nova (*New map of Africa*)

| | |
|---|---|
| **Artist** | Abraham Ortelius |
| **Place** | Antwerp |
| **Year** | 1570 |
| **Dimensions** | 37.5 × 51 cm (14.8 × 20.1 in.) |
| **Location** | Louis O'Neill Collection |

This hand-colored plate is the first map of the entire African continent to appear in an atlas. Like every map in Ortelius's *Theatrum Orbis Terrarum* ("Theater of the World"), it is accompanied by descriptive text for a curious European audience, in which Ortelius tells readers that key information about the continent came from an African-born diplomat named Leo Africanus. Notes on the map itself offer additional explanations; a text block in the lower right reads, "This island is called Madagascar by its inhabitants, St. Lawrence by the Spaniards; formerly Menuthias by Ptolemy, perhaps Cerne by Pliny." Lakes and rivers are simplified and inaccurate, reflecting incomplete European knowledge, while sea monsters and ships in a storm allude to the captivating drama of maritime travel.




# COAST of AFRICA

*from the*

*STRAITS of GIBRALTAR to CABO VERDE.*

*drawn from Various Geographical M.S.S.*

*by A. Arrowsmith.*

Published 2nd Aug.ᵗ 1804 by Cadell & Davies Strand.

# 20. Coast of Africa from the Straits of Gibraltar to Cabo Verde

| | |
|---|---|
| **Artist** | Aaron Arrowsmith |
| **Place** | London |
| **Year** | 1803 |
| **Dimensions** | 40 × 25.5 cm (15.7 × 10 in.) |
| **Location** | Dr. Oscar I. Norwich collection of maps of Africa and its islands |

English mapmaker Aaron Arrowsmith drew this chart for a set of five to accompany James Stanier Clark's *The Progress of Maritime Discovery*, a study of early navigational methods and naval history meant for tracing voyages from an armchair and not from a ship at sea. Its title reveals his method: "drawn from various geographical MSS," or manuscripts, and the mixture shows in place names written in Spanish, Portuguese, English, and Arabic along the coast. Soundings, obstructions, ports, and river mouths dot the shoreline of what is now Morocco and Western Sahara while the interior appears nearly empty, highlighting the utility of this map for understanding maritime routes. The map leaves out one element that would typically be found on maps made for navigation: headland views showing the profile of land from the water used to orient oneself offshore. Their absence is evidence this map wasn't meant to be used as a practical navigational aid, but instead as a record of the voyages Clark's book describes.

WEST AFRICA.
PARK'S DISCOVERIES
EXPLANATION.
Edinburgh, Published by Wingh & Innes, & Kirkwood & Son, 1820.
Kirkwood's Sculp.
1820

# 21. West Africa

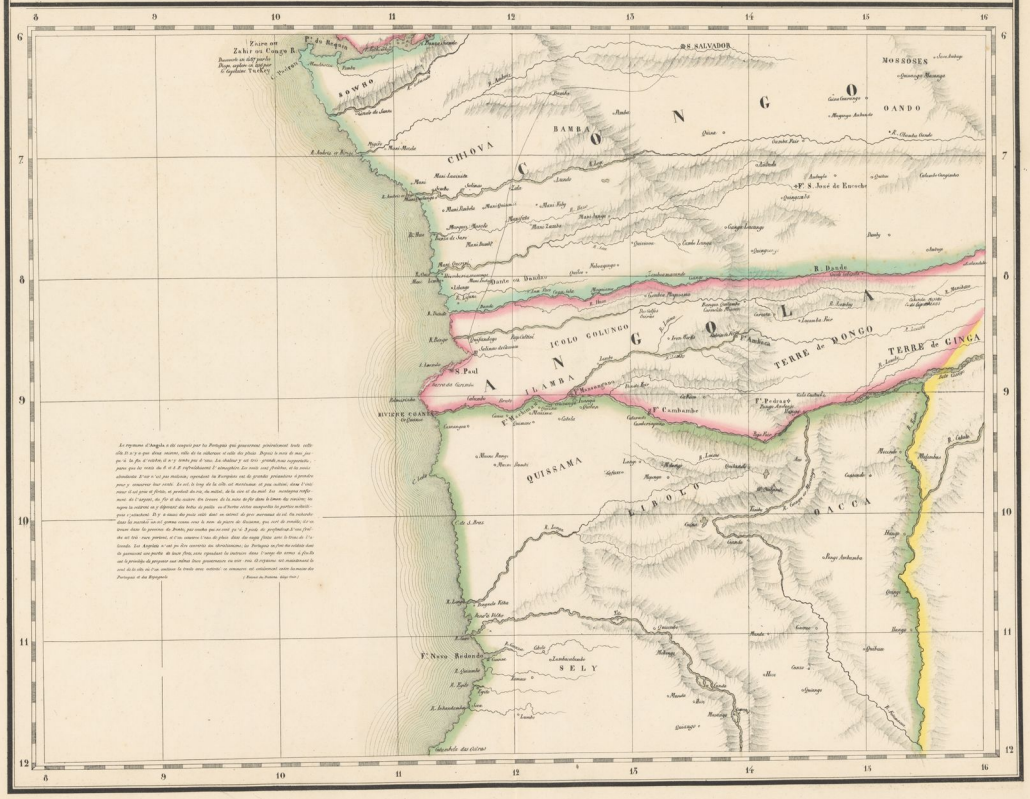
| | |
|---|---|
| **Artist** | Kirkwood & Sons |
| **Place** | Edinburgh |
| **Year** | 1820 |
| **Dimensions** | 33 × 27 cm (13 × 10.6 in.) |

By the 1820s, European fascination with Africa had shifted from coastlines to the interior, and the story of Mungo Park, the Scottish explorer of the Niger River, captivated the public. This map highlights his celebrity status across the continent's middle, where "Park's Discoveries" draws attention to his 1795-96 route out and return in red, as well as his fatal 1805-06 expedition in blue. Park is believed to have drowned at the Niger's Bussa rapids after a conflict with local people, but the account was met with some skepticism in Europe, and the mystery of his fate was still alive when this map was published. Around Park's routes the map shows what Europeans knew, or thought they knew, about the interior: deserts, quicksand, gum forests, caravan tracks dotted with settlements, and the territories of the region's many peoples.

AFRIQUE.

# CONGO ET ANGOLA.

N°. 41.


Zaire ou Zahir ou Congo R.
Pool de Malebo
Embouchure du Zaire
SOUNDO
CHILOVA
BAMBA
NGOLA
S.S SALVADOR
NOSSOSES
OANDO
Mont Lemba
Mont Ambriz
Mont Mocolo
Mont Zambe
Mont Bondé
Mont Bailundo
Mont Bailé
Mont Quissama
Mont Cavalle
Mont Cavalla
Mont Cavalle
Mont Cavalla
Mont Cavalla
Mont Cavalla

# 22. Afrique: Congo et Angola (*Africa: Congo and Angola*)

| | |
|---|---|
| **Artist** | Philippe Vandermaelen |
| **Place** | Brussels |
| **Year** | 1827 |
| **Dimensions** | 37 × 53 cm (14.6 × 20.9 in.) |
| **Location** | Louis O'Neill Collection |

Vandermaelen's six-volume atlas, *Atlas universel de géographie physique, politique, statistique et minéralogique* ("Universal atlas of physical, political, statistical and mineralogical Geography"), with one volume per inhabited continent, was the first to map the entire world at a uniform scale. In volume three, Africa is divided not into the political or cultural regions Europeans imagined for it, but into standard numbered sections at a constant scale with a consistent set of symbols. The scale allows the maps to be detailed enough to show more precise settlements, river features, and caravan routes than the maps before it. At first, this seems a more objective way to map a continent, but the impression breaks down in the descriptive text. Printed on this sheet in French, it opens with the Portuguese conquest of Angola, inventories the area's extractable resources, and measures the climate based on what Europeans could endure. The frame of the map was standardized, but only served to emphasize a European point of view.

TRANSVAAL
ORANGE RIVER COLONY
CAPE COLONY
NATAL
INDIAN OCEAN
ORANGE RIVER
Mafeking
Pretoria
Johannesburg
Kroonstad
Bloemfontein
Kimberley
Colesberg
Cape Town
Port Elizabeth
Durban
East London
Port Alfred
Newcastle
Ladysmith
Port Natal
Harrismith
Aliwal North
Port Beaufort
Cape Agulhas
Simons Bay
Cape or Good Hope
Wynberg
St Helena Bay
Lord Roberts
Lord Kitchener
Sir Redvers Buller
Sir George White
Sir Penn Symons
H.M. THE QUEEN
DESIGNED AND PUBLISHED BY M. HENRY HARGREAVES, CHORLTON-CUM-HARDY, MANCHESTER, ENGLAND.

# 23. [South Africa]

| | |
|---|---|
| **Artist** | M. Henry Hargreaves |
| **Place** | Manchester |
| **Year** | 1900 |
| **Dimensions** | 63 × 93 cm (24.8 × 36.6 in.) |
| **Location** | David Rumsey Map Collection |

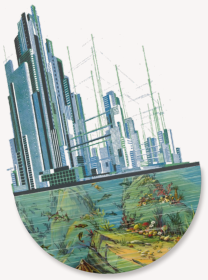
The only known copy of this untitled pictorial map shows Southern Africa in the midst of the Second Boer War, fought between the British Empire and the Boer settler republics of the Transvaal and Orange Free State over African land and its rich gold resources. Though remembered as a war between white powers, Black South Africans served and suffered in it by the tens of thousands. The map itself is a patriotic muscle-flex for the British public, showing Queen Victoria at the top and center, with commanders Roberts and Kitchener, fresh from victories in the top left corner. Bird's-eye views of Cape Town, Johannesburg, and other settlements dot the vivid green landscape which conspicuously contains no representation of indigenous inhabitants.

# 24. Iwo-oorun Alkebu-lan (*West Africa*)

| | |
|---|---|
| **Artist** | Nikolaj Jesper Cyon |
| **Place** | Stockholm |
| **Year** | 2024 |
| **Dimensions** | 63 × 92 cm (24.8 × 36.2 in.) |

What would Africa look like if Europe had never colonized the continent? Swedish artist Nikolaj Cyon asks us to imagine exactly that in this map of West Africa as it might have appeared in 1853, a generation before the Berlin Conference carved up the continent between European powers. Nearly every cartographic convention represented elsewhere in this section is deliberately challenged. South is oriented at the top, places are labeled with local endonyms rather than European placenames, and the title, written in Yoruba, translates as "Western Alkebu-lan," a reference to what some scholars believe to be an indigenous pre-colonial name for Africa.

# Data Visualization Beyond the Map

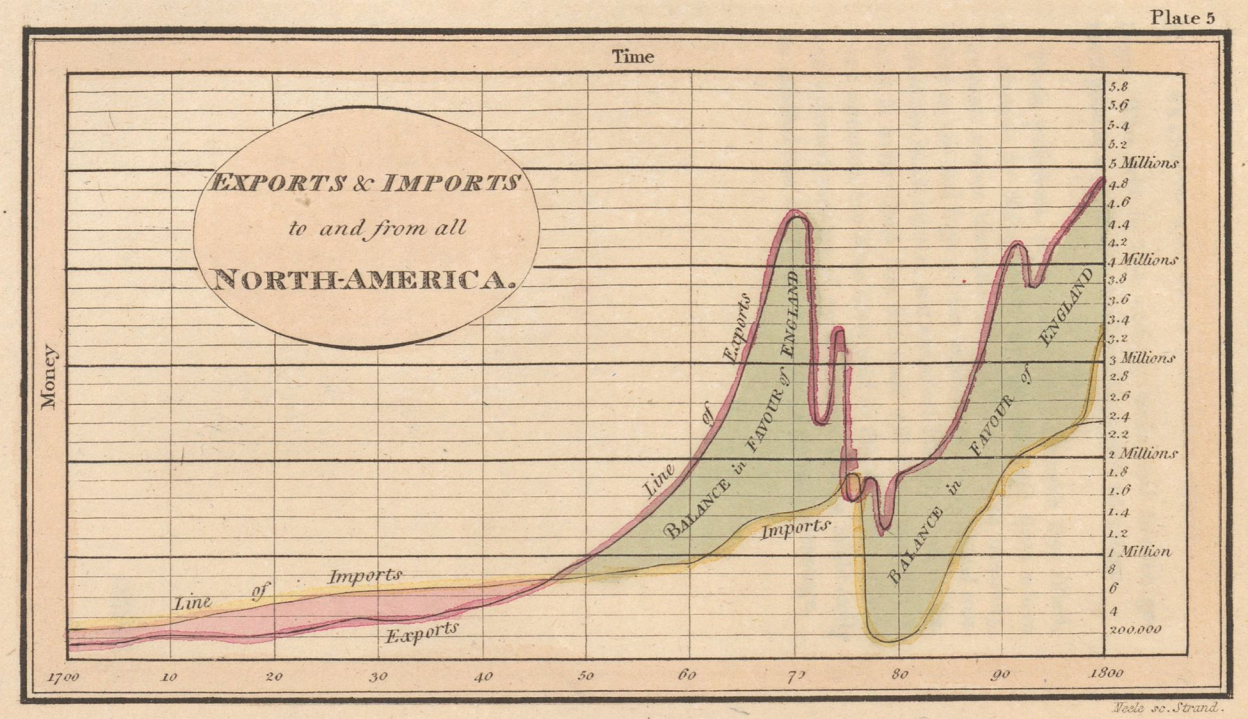


Communicating information is at the very core of cartographic expression, yet maps are not the only visual resource type the Center holds. Our collection of historical data graphics presents statistics and other information through many of the same visual techniques as maps, but without relying on geographic space to make a point or tell a story.

The pioneers of data graphics appear throughout this collection. William Playfair invented the line graph and bar chart in the 1780s, “painting to the eye” a century of British and North American trade in one view. In the 1920s, Otto and Marie Neurath, with artist Gerd Arnz, developed the International System of Typographic Picture Education, the Isotype, which uses rows of repeated pictograms to make social and economic statistics visually appealing, memorable, and easy to understand. This visual language spread quickly across Europe and the rest of the world, and is still used in the work of designers today as evident in the graphics that follow.

The Center embraces a wide ranging definition of cartography, stemming from our commitment to enhance learning in visual ways whether or not they are place-based. Historical data graphics expand how we teach deep looking at visual material of every kind, and provide a unique lens through which we can see history in a simple and approachable way.

Plate 5

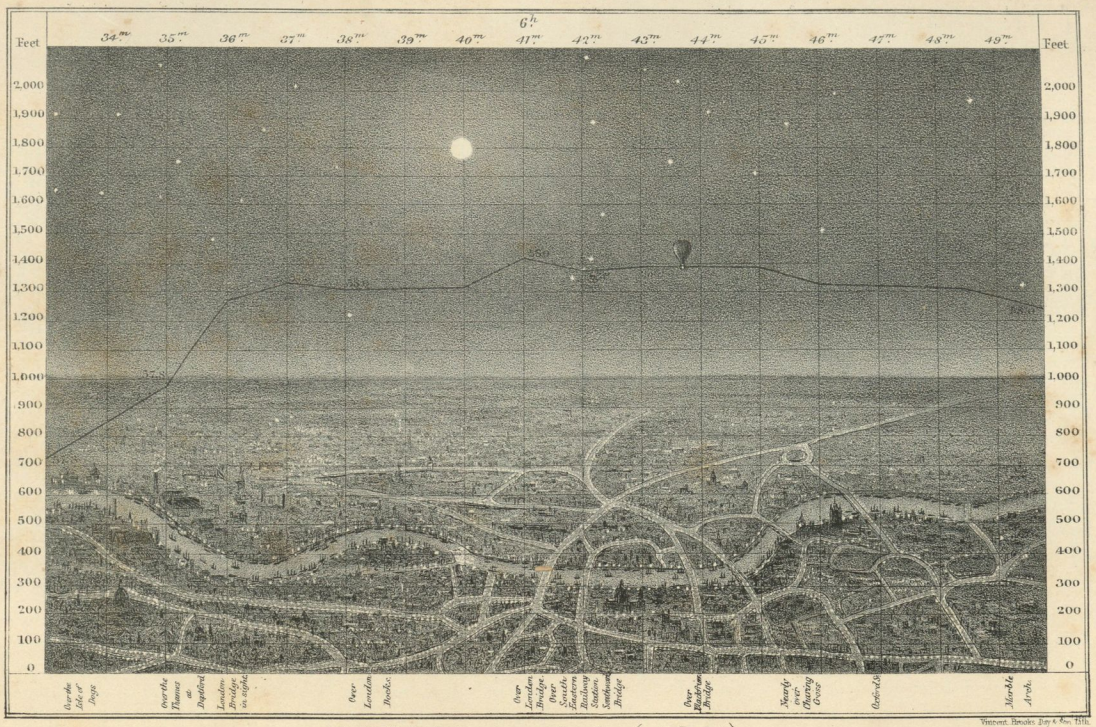



Neele sc. Strand.

# 25. Exports & Imports to and from all North - America.

| | |
|---|---|
| **Artist** | William Playfair |
| **Place** | London |
| **Year** | 1801 |
| **Dimensions** | 10 × 17 cm (3.9 × 6.7 in.) |
| **Location** | David Rumsey Map Collection |

In the late 18th century, William Playfair invented many aspects of the visual language we use to communicate with data today. The line graph and bar chart first appeared in his 1786 *Commercial and Political Atlas*, an atlas containing no geographic maps at all, stretching the word to mean a compendium of information presented visually. On this chart, a century of British trade with North America is shown. The gap shaded with green between the pink export and yellow import lines mark the “balance in favour of England,” which pinches abruptly to nothing as the American Revolution halts trade, then rebounds again after peace. In the pages of observations regarding trade with North America, Playfair supported his new visual method as “painting to the eye” to show lines representing quantity and time “with mathematical exactness that is not to be called in question.”

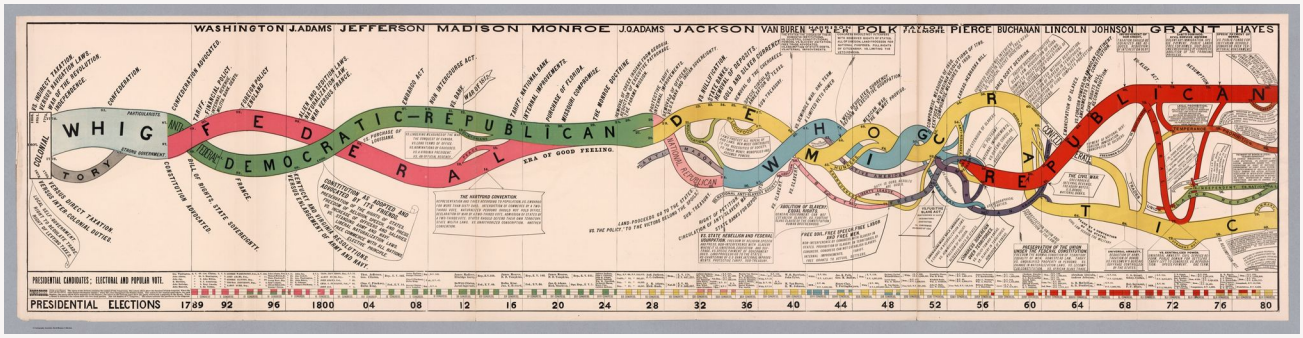



Path of the Balloon over London, (at Night.)
2nd. Octr. 1865.

# 26. Path of the Balloon over London (at Night)

| | |
|---|---|
| **Artist** | James Glaisher |
| **Place** | London |
| **Year** | 1871 |
| **Dimensions** | 18 × 27 cm (7.1 × 10.6 in.) |
| **Location** | David Rumsey Map Collection |

Victorian-era ballooning turned the sky into a place of scientific inquiry, and James Glaisher, exploring what he called the “aerial ocean,” worked to demystify its “depths for the causes of atmospheric phenomena.” This plate is a chart, a map, and a view all at once. The balloon’s altitude is plotted minute-by-minute as a line set over a moonlit view of London which appears to float over the streets below. On this particular flight, Glaisher recorded something he felt “remarkable”: the air temperature rose as did the balloon, an inversion of his expectations. *Travels in the Air* contains 125 illustrations of awe-inspiring sights, and serves as an example of data visualization as a thing of beauty, a visual gift presented to readers who would themselves never leave the ground.


WASHINGTON J.ADAMS JEFFERSON MADISON MONROE J.QADAMS JACKSON VAN BUREN HARRISON POLK TAYLOR PIERCE BUCHANAN LINCOLN JOHNSON GRANT HAYES
WHIG
TORY
FEDERAL
DEMOCRATIC-REPUBLICAN
DEMOCRATIC
WHIG
REPUBLICAN
ERA OF GOOD FEELING.
PRESIDENTIAL CANDIDATES; ELECTORAL AND POPULAR VOTE.
PRESIDENTIAL ELECTIONS 1789 92 96 1800 04 08 12 16 20 24 28 32 36 40 44 48 52 56 60 64 68 72 76 80

# 27. Presidential Elections

| | |
|---|---|
| **Artist** | Walter Houghton |
| **Place** | Indianapolis |
| **Year** | 1880 |
| **Dimensions** | 34 × 142 cm (13.4 × 55.9 in.) |
| **Location** | David Rumsey Map Collection |

Here the history of the American political system is depicted like a winding river. Parties of the first century of American history flow, merge, and split into tributaries, with all of the turbulence visible in one view. Color represents different political parties, width shows their strength, and the one holding power is positioned on top. The current smooths into one broad stream during the “Era of Good Feeling” where parties merged after 1817, and what follows twists into rapids during the decades leading up to the American Civil War. Houghton’s accomplishment in his chart, included in *Conspectus of The History of Political Parties and the Federal Government*, was to provide clarity at a glance about an important history impossible to memorize and hard to find summarized anywhere.

# INDUSTRIA DELLE CONSERVE ALIMENTARI

## ESPORTAZIONE DI CONSERVE DI POMODORO (QL.)

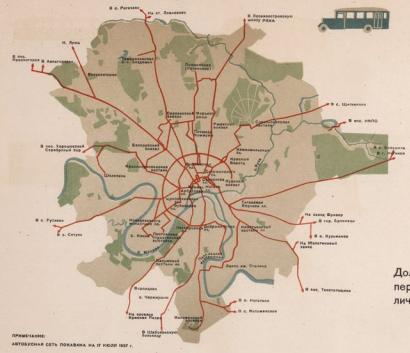
## ESPORTAZIONE DI CARNI SALATE AFFUMICATE O IN ALTRO MODO PREPARATE (ESCLUSO IL LARDO) (QL.)

# 28. Industria delle conserve alimentari (*Food canning industry*)

| | |
|---|---|
| **Artist** | Confederazione generale dell'industria italiana |
| **Place** | Rome |
| **Year** | 1929 |
| **Dimensions** | 22 × 32 cm (8.7 × 12.6 in.) |
| **Location** | David Rumsey Map Collection |

Maps and data visualizations work together in *Lo sviluppo dell'industria italiana* ("The Development of Italian Industry"), a thematic national atlas of 109 charts and three maps tracing growth from 1876 to 1928. In the atlas, pictorial statistics are given a distinctly Italian flavor. On this plate, devoted to the food preserving industry, exports of tomato preserves grow into larger cans, while exports of salted and smoked meats, "lard excluded," hang on lengthening racks of prosciutto and salami. The delight is real, and the messaging is memorable. Published under Mussolini's regime by Italy's industrial confederation, this atlas represents a nation that only grows: statistics working with the national self-portrait during Italy's early Fascist period.

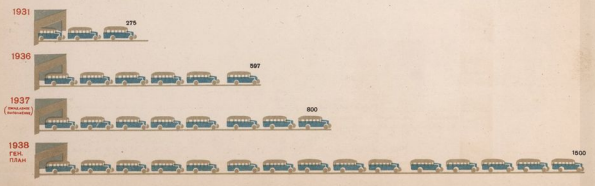
## АВТОБУСНАЯ СЕТЬ



ПРИМЕЧАНИЕ:
АВТОБУСНАЯ СЕТЬ ПОКАЗАНА НА 17 ИЮЛЯ 1937 г.

Доля автобуса в московских перевозках из года в год увеличивается

## АВТОБУСНЫЙ ПАРК (ИНВЕНТАРНОЕ ЧИСЛО АВТОБУСОВ)

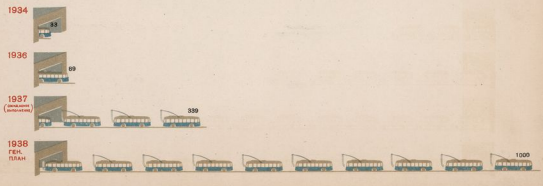



## ТРОЛЛЕЙБУСНАЯ СЕТЬ

*** ПРОЕКТИРУЕМЫЕ ЛИНИИ ТРОЛЛЕЙБУСА НА 1938 Г. (ВО ТРЕТЬЕМ КВАРТАЛЕ)
ДЕЙСТВУЮЩИЕ ЛИНИИ ТРОЛЛЕЙБУСА НА 1 ЯНВАРЯ 1938 Г.

Троллейбус сохраняет чистоту воздуха города, поэтому троллейбусный парк в новой Москве быстро увеличивается

## ТРОЛЛЕЙБУСНЫЙ ПАРК (ИНВЕНТАРНОЕ ЧИСЛО ТРОЛЛЕЙБУСОВ)



# 29. Left: АВТОБУСНАЯ СЕТЬ (*Bus network*); right: ТРОЛЛЕЙБУСНАЯ СЕТЬ (*Trolleybus network*)

| | |
|---|---|
| **Artist** | Vsesoiuznyĭ nauchno-izdatel'skiĭ institut izobrazitel'noĭ statistiki (*All-Union Scientific and Publishing Institute of Pictorial Statistics*) |
| **Place** | Moscow |
| **Year** | 1938 |
| **Dimensions** | 35 × 35 cm (13.8 × 13.8 in.) |
| **Location** | David Rumsey Map Collection |

Innovations of the Neuraths' Isotype system are demonstrated here in the use of repeating units to show a numerical increase (as opposed to scaling the size of a pictogram, as in the adjacent Italian atlas) and elegant fusion of statistical accuracy with simplified representational elements. In *МОСКВА РЕКОНСТРУИРУЕТСЯ — All-union Institute of Pictorial Statistics of Soviet Construction and Economy*, this Soviet planning atlas celebrates Stalin's reconstruction of Moscow with visual simplicity and clarity. These pages show growth in the city's number of buses and trolleys in the years leading up to 1938. The maps show the existing networks these modes of transportation operated, with the dotted line in the map on the right representing improvements projected for 1938. The quote below the map reads "The trolleybus keeps the city's air clean; therefore, the trolleybus fleet in the new Moscow is increasing rapidly." The full atlas's 260 pages of persuasive civil-planning imagery and before-and-after evidence of the progress made in Stalin's Moscow.

# 30. Cross Sections through California

| | |
|---|---|
| **Artist** | RJ Andrews |
| **Place** | San Francisco |
| **Year** | 2020 |
| **Dimensions** | 72 × 25 cm (28.3 × 9.8 in.) |
| **Location** | David Rumsey Map Collection |

Data storyteller RJ Andrews quotes Otto Neurath in his *Cross Sections Through California*: "The purpose of teaching-pictures is to have an effect on the mind." The historical styles from this section are alive and working to slice up California into seventeen elevation profiles, their land cover summarized in Isotype-like symbols, explicitly honoring the system Otto and Marie Neurath and Gerd Arntz invented a century ago, such as sans serif lettering, pictograms, and color applied sparingly but with purpose. His choices are as cartographic as they are graphic, rotating north to fit the state into the frame the way we imagine it, valleys and mountains going up and down, with the coast facing out to the Pacific. On the reverse, pictogram charts count crops, electricity, and population since 1850, employing symbols the Isotype pioneers never needed, like solar panels, and honest footnotes, such as that the 1850 census counted only seven thousand women and mostly omitted the state's Indigenous population.

# Höhen Charte

oder

bildlich vergleichende Übersicht der bedeutendsten Berge

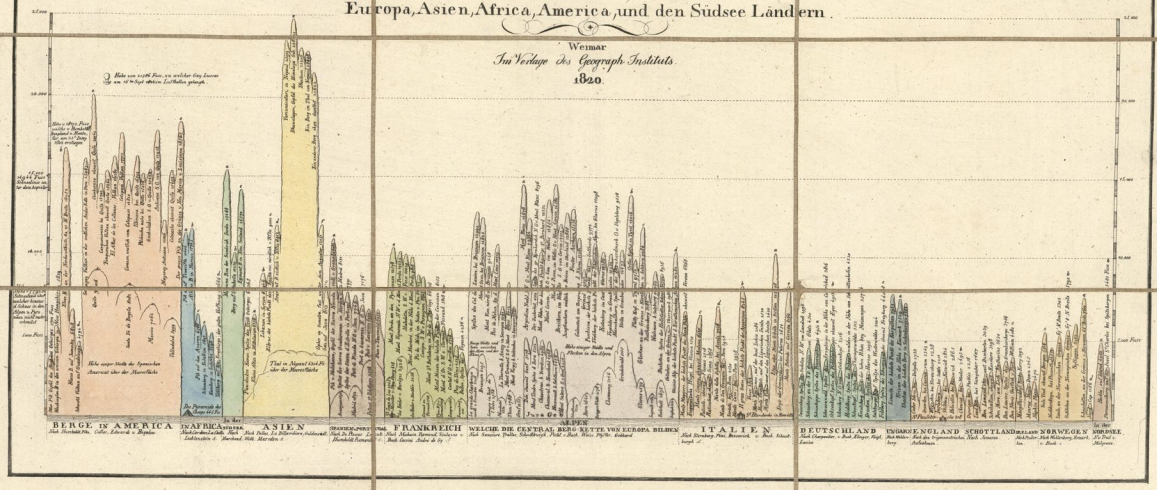
in

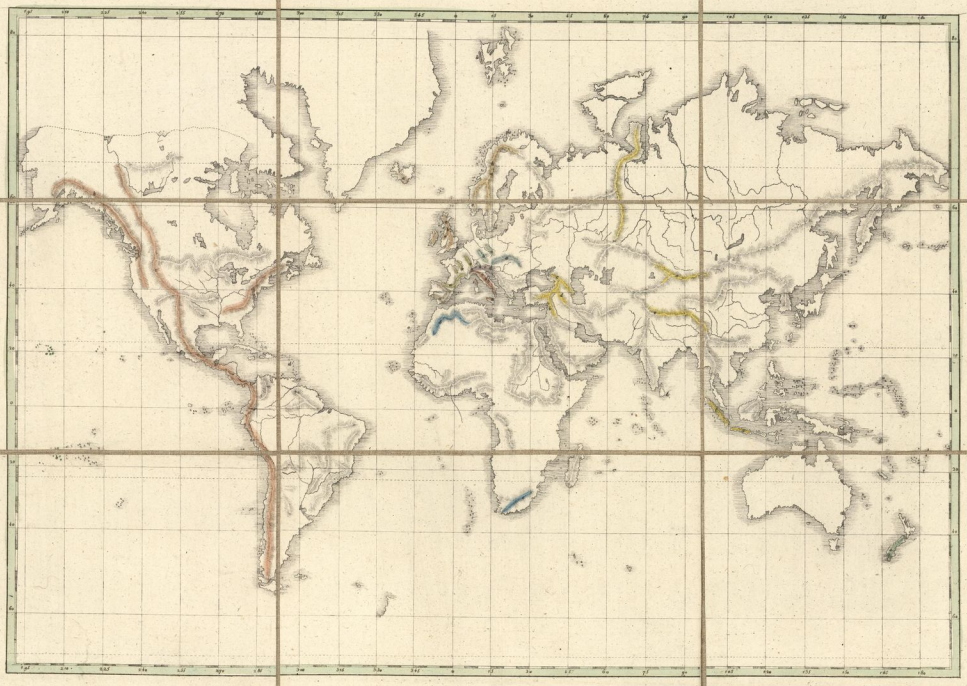
Europa, Asien, Africa, America, und den Südsee Ländern.

Weimar

Im Verlage des Geograph. Instituts.

1820.



## 31. Hohen Charte oder bildlich vergleichende Ubersicht der bedeutendsten Berge (*Elevational Chart, or a Pictorial Comparative Overview of the Most Significant Mountains*)

| | |
|---|---|
| **Artist** | Carl Ferdinand Weiland |
| **Place** | Weimar |
| **Year** | 1820 |
| **Dimensions** | 63 × 50 cm (24.8 × 19.7 in.) |
| **Location** | David Rumsey Map Collection |

Pairing a chart with a map enables each to add value to the other. The chart at the top shows the heights of major mountains, and the map shows where they are. Ranges are color-coded to correspond with the map below, allowing readers to move easily between the two. Notable world landmarks are shown to give a sense of scale, such as the Great Pyramid of Giza at the base of the blue Africa section. At the top of the scale is the flight altitude reached by Louis Gay-Lussac on his 1804 balloon flight. The design borrows from Alexander von Humboldt's earlier data-driven profiles of the Andes, which helped launch a new era of visualizing geographic phenomena. Comparative mountain charts became a common visualization in geography publications in the decades that followed. Many of these showed crowded mountain peaks in one imagined landscape, but Weiland instead arranges them by mountain range along a measurable scale.

# Unconventional Cartography: Counter-mapping and Artists' Maps

Mapmaking takes many forms, from artists interpreting their own vision of the world, citizens mapping for social change without regard for cartographic conventions, and practitioners pushing cartography in entirely new directions. This collection is largely unconstrained by traditional definitions of cartography or collectibility, making it a unique corpus of map-ish materials.

These works stretch the imagination of how to see the world. Projections abandon the globe for a pyramid. A neighborhood becomes the center stage. Toxic sites bleed through a landscape like dripping black ink. Each one asks us to think beyond the conventions that the rest of our collections helped establish.

They also open the practice itself. Cartography has traditionally lived inside institutions, with its tools and authority held by surveyors, publishers, and governments. The map makers in this section claim those tools for everyone and treat the map as personal expression, protest, and fun, making cartography an art for all.

# 32. [Map Projections]

| | |
|---|---|
| **Artist** | Agnes Denes |
| **Place** | Cambridge |
| **Year** | 1978 |
| **Dimensions** | 28 × 21 cm (11 × 8.3 in.) |
| **Location** | David Rumsey Map Collection |

In this piece cartography becomes an artistic language that makes distortion the purpose. Agnes Denes, a leading pioneer of environmental and conceptual art, spent the 1970s asking versions of that question in her series *Isometric Systems in Isotropic Space: Map Projections*, which she completed just before this retrospective exhibition at MIT's Hayden Gallery. This catalog cover, printed on a graph-paper grid, summarizes the project well. What if Earth were not a sphere? Shown here are six pyramids, three with the continents projected on them, while the others show the globe's latitude and longitude grid lines stretched over the same unexpected form. Denes projected the world onto pyramids, eggs, snail shells, and hot dogs using the mathematics of map projections, which always distort, to challenge any map's claim to objectivity.

MAP
of
GREENWICH
VILLAGE
made for
THE WHITNEY
STUDIO CLUB

# 33. Map of Greenwich Village made for the Whitney Studio Club

| | |
|---|---|
| **Artist** | Kaj Klitgaard and William Edwin Rudge II |
| **Place** | New York |
| **Year** | 1920 |
| **Dimensions** | 59 × 75 cm (23.2 × 29.5 in.) |
| **Location** | David Rumsey Map Collection |

Kaj Klitgaard uses old cartographic conventions with an artist's eye. The Whitney Studio Club, founded by sculptor Gertrude Vanderbilt Whitney in 1914, was a haven for Greenwich Village painters and students, and mounted at least ten exhibitions per season featuring unknown artists. This map, commissioned by the Whitney Studio Club, was a map of artists, by artists, for artists. He adds a mermaid and sea monster offshore, includes a scroll cartouche, and renders the neighborhood a miniature model seen from a bird's-eye view. But he also holds these rules loosely, with depth emphasized by clouds and a magnifying glass hovering over the club itself, allowing the viewer to peer inside. Around the border, portraits of Village artists and verses complete the spirit of the Village's modern art movement.

# 34. Map to Eric’s Place in Muir Beach

| | |
|---|---|
| **Artist** | Robert Crumb |
| **Year** | 1972 |
| **Dimensions** | 36 × 28 cm (14.2 × 11 in.) |
| **Location** | David Rumsey Map Collection |

Hand-drawn on the back cover of a spiral-bound notepad, this map was not for commission or press, but to capture directions to the house of a friend. Artist and cartoonist Robert Crumb, perhaps best known for work on late 20th century underground comics series like Zap and characters like Mr. Natural, has traced his personal route from San Francisco, over the Golden Gate Bridge, to Muir Beach, noting the landmarks his memory actually keeps: Richfield station, a row of mailboxes, and a rope fence. Yet the mapmaker’s instincts are all there, like the compass rose, an old Spanish galleon sailing offshore, a vignette of Eric’s house above the water, all shown with a nod to any antique map. It’s not a polished piece, but a glimpse into the artist’s mind. Nonetheless, it does what all maps do: illustrate the world from a particular point of view.

# 35. A Cultural Map: Southeast San Francisco

| | |
|---|---|
| **Artist** | Kate Connell, Oscar Melara, and Sofía Vivanco Airaghi |
| **Place** | San Francisco |
| **Year** | 2017 |
| **Dimensions** | 30 × 40 cm (11.8 × 15.7 in.) |
| **Location** | David Rumsey Map Collection |

*A Cultural Map: Southeast San Francisco* was conceived and designed by artists and is populated with places of importance to neighborhood residents, and its unconventional style of cartography measures a place by what its citizens care about most. Its contents were gathered during an exhibition at City College of San Francisco, labeled here just west of the Excelsior neighborhood, where visitors shared their ideas for what belonged on the map. Contributors' names appear within the green outer border. The result is a subjective geography of the area: less of a record of the physical landscape than a portrait of community values, made by citizens, for citizens.

# 36. Swamp then lake now

| | |
|---|---|
| **Artist** | Veronica Graham |
| **Place** | Oakland |
| **Year** | 2023 |
| **Dimensions** | 40.8 × 27 cm (16.1 × 10.6 in.) |

This silkscreen-printed, two-part abstracted view of Lake Merritt—a public space at the heart of Oakland's culture and lifestyle—pays homage to the ways illustrative maps have long told the stories of a place. The print reads in two parts using symbols and patterns to evoke the landscape: the lower half represents the area before European settlers arrived, while the upper half shows the lake as it appears today, surrounded by red buildings, homes, and other industrial symbols. Graham illustrates this local environment much as a cartographer would, using color, space, scale, and the relationships between objects to tell as visual, place-based history. Maps can help us to understand how landscapes have changed over time, just as Graham has done in one glimpse.

# 37. Land of Opportunity

| | |
|---|---|
| **Artist** | Connie Zheng |
| **Place** | San Francisco |
| **Year** | 2022 |
| **Dimensions** | 213 × 213 cm (83.9 × 83.9 in.) |
| **Location** | David Rumsey Map Collection |

In this large mixed-media piece Connie Zheng maps forty-three of the most notorious toxic sites in the San Francisco Bay Area with oozing, tar-like blobs symbolizing ground poisoned by high-tech manufacturing, military sites, and gold-rush era mining. This is thematic mapping as an artist practices it, through symbolism, juxtaposition, and emotion rather than statistical convention. Its handwritten notes record protests, lawsuits, and remediation efforts by which communities have fought for a healthier environment. Zheng calls this work “a register of the ways in which the Bay Area has been shaped, both figuratively and literally, by dominant settler narratives of the landscape’s potential for wealth and resource extraction.” Is it a land of opportunity, all things considered? The map asks us to weigh the answer.

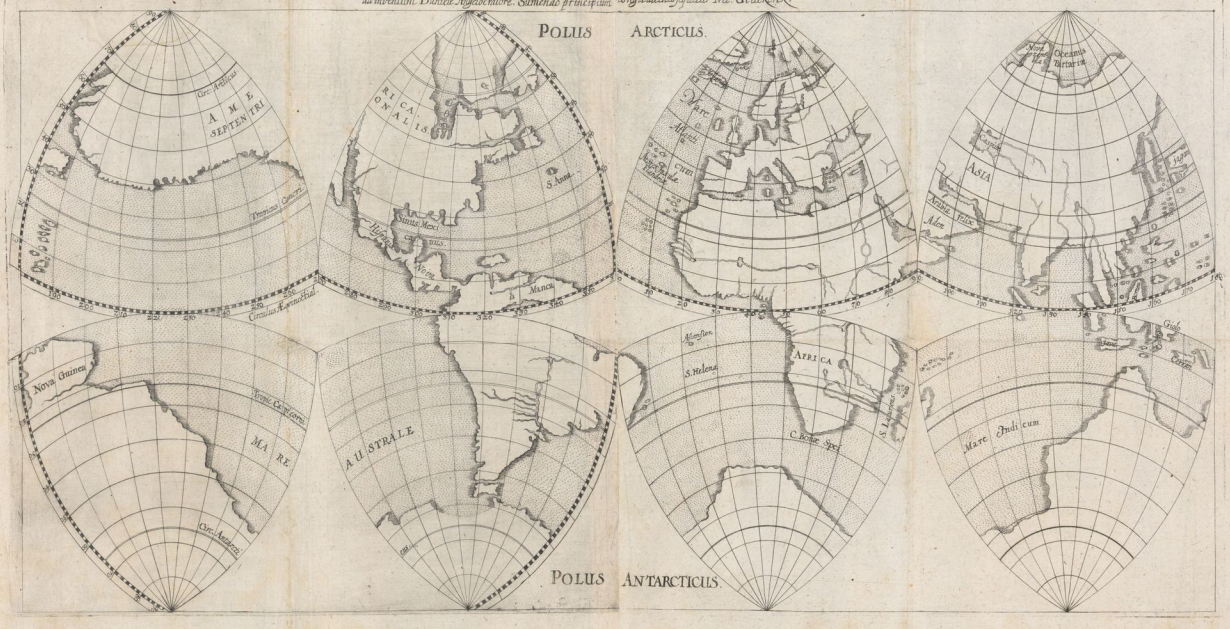
# Measuring and Mapping the World

Maps and the people who made them have shaped how we understand the world, and that influence can be especially appreciated through the evolution of the Earth-measurement sciences and technology behind them. This collection looks at the history of geomatics: the science and technology behind the collecting, analyzing, and presenting spatial information. The term is modern, but it has a history starting in the analog era, represented by items such as compasses, surveying instruments, field notebooks, and map makers' tools.

Before satellites and geographic information systems (GIS), every coastline was traced by a navigator positioning themself from celestial objects, every mountain measured by surveyors' angles, and every accurate map built from the thousands of collective observations made by hand, recorded in notebooks, and passed on to new generations of mapmakers.

Where other collections at the Center consider maps as finished products, this one looks under the hood. Its objects, projections, declination charts, ship's logs, field notebooks, compasses, and other sighting instruments are the working tools of measurement. They represent the analog versions of aerial photography, satellite navigation, and GIS. They let us understand the history of our evolving ability to measure the Earth, and to appreciate how much labor and ingenuity is underneath the maps in our collections.

NOVUM ORBIS TERRARUM SCHEMA, IN PLANO SIC DESCRIPTUM, UT GRADUUM TUM CIRCA POLOS DILATATIO, TUM CIRCA AQUATOREM COARCTATIO EXCLUDATUR,
ad inventum Danielis Aygelocentoris. Sumendo principium longitudinis sequitur Nic. Gerlekerck.



# 38. Novum orbis terrarum schema, in plano sic descriptum, ut graduumtum circa polos dilatatio, tum circa aequatorem coarctatio excludatur (*A new diagram of the world, drawn on a flat surface in such a way that both the stretching of degrees near the poles and the crowding of degrees near the equator are avoided*)

| | |
|---|---|
| **Artist** | Daniel Angelocrator |
| **Place** | Frankfurt |
| **Year** | 1628 |
| **Dimensions** | 31.9 × 55.6 cm (12.6 × 21.9 in.) |

From early on, cartographers grappled with the problem of depicting the three-dimensional earth on a two-dimensional surface. As European exploration expanded their knowledge of the world, mapmakers experimented with ways of fitting the globe on a single sheet of paper so that newly encountered lands could be seen in context. World maps showing the earth as two hemispheres were common in this era, but omitted or distorted regions near their edges. Angelocrator's solution in this map was to split the sphere into eight sections, like slicing an apple in half and in half again to get eight equal parts, instead of one or two, reducing the apparent "stretching" of landforms. Shown here are the eight sections arranged together to depict the whole earth. Angelocrator also created another arrangement depicting the earth from its poles as two maps of the northern and southern hemispheres. Each of the eight sections take the form of what we now call a Reuleaux triangle, a curved shape of constant width that Leonardo da Vinci used a century earlier for his own experimental world map.

Mer Du Nord
Mer du Sud ou Pacifique
Essay dune Nouuelle Carte de L'Amerique Meridionale.
Equateur ou Ligne Equinoxiale
Tropique du Capricorne
Tropique de L'Ecrevisse
Terre Ferme
Echelle de Longitude Occidentale du Meridien de Paris
Echelle de Longitude Orientale du Meridien de Teneriffe

# 39. Essay d’une nouvelle carte de L’Amerique Meridionale (*Draft of a new map of South America*)

| | |
|---|---|
| **Artist** | Author unknown |
| **Year** | 1740 |
| **Dimensions** | 88.8 × 56.2 cm (35 × 22.1 in.) |

This hand-drawn chart of South America demonstrates how cartographers critiqued, refined, and corrected existing maps as better information became available to them. This map records an unknown cartographer’s efforts to compare the continent’s coastline as shown on existing charts, represented by the thin line off the coast of Brazil, with a new outline informed by fourteen years of improved astronomical observations. Curved red lines in the South Atlantic represent changes in compass variation caused by the Earth’s magnetic field. Today, mariners use sophisticated data models to calculate magnetic declination, or the difference between true north and where the compass points. At the time of this map’s creation, navigators and scientists were working out methods for predicting and mapping this variation using specialized instruments.

# SYNOPSIS PLATE.

SHEWING SOME OF THE PRINCIPAL CHARACTERS USED IN MAPPING.

# 6. Synopsis plate shewing some of the principal characters used in mapping

| | |
|---|---|
| **Artist** | Benjamin P. Wilme |
| **Place** | London |
| **Year** | 1841 |
| **Dimensions** | 22 × 28 cm (8.7 × 11 in.) |
| **Location** | David Rumsey Map Collection |

The rapidly industrializing 19th-century world ushered in a movement toward standardization across many fields. Guides such as *A Hand book for plain and ornamental mapping, and engineering drawing* helped establish shared cartographic conventions. Wilme, a British civil engineer and surveyor, compiled a utilitarian reference with a distinctly educational purpose: to use dozens of plates and fold-outs to show how a well-designed engraved map should be made. Its pages offer advice on lettering styles; methods for representing trees, sand, water, and mountains; and instructions for using essential drawing tools such as stencils. It was meant to both teach aspiring mapmakers, architects, and engineers, and to serve those already professionally involved in efforts by the Ordnance Survey, Britain's national mapping agency, and the booming railway industry. This plate gathers "some of the principal characters used in mapping" into a single imagined landscape, such as hilly landscapes, fortifications, gardens, and land use—some of which are evident in maps on display here.

H. M. S. "Swiftsure" From Esquimalt to San Francisco NOV 1884

| H. | M. | S. | Remarks |
|---|---|---|---|
| 1 | 7 | 0 | Rec. |
| | | | 8.50 Made plain sail less Mainsail & J jib |
| | | | Empd scrubbing Canoe gear |
| | | | 11.0. Shortened and furled sails pointed yards to the wind |
| | | | 12-0-0 Log 128 |
| | | | Made good. South 120. |
| | | | Cape Mendocino S. 50 E. 180. |
| | | | Rec |
| | | | Empd as requisite |
| | | | 4.0. Land on Port beam. |
| | | | 4.10. as often. Read bearings 235 |
| | | | Made plain sail. Shortened & furled sails. pointed yards to wind Sqr Fs a Sails |
| | | | 12—0—0. |
| | | | Rec |
| | | | 12.20. Ex fore and aft sails |
| | | | 4.20. Inspected Ships companies Read prayers. Read Instructions Station for. Securing plain sail. Shortening do. & shifting top gt masts. bowse top gt masts. |
| | | | Read Instructions for Watch close watertight doors |
| | | | 12—0—0 |
| | | | Made good. S. 2 E. 126 miles |
| | | | Cape Mendocino S 10 E. 214 miles |
| | | | Rec |
| | | | Empd as requisite |
| | | | 3.20. Read Instructions for watch close watertight doors |
| | | | 4.10 as often Sent fair stations |
| | | | 8.0. Log 252 Miles |
| | | | 12—0—0. |

# 40. Log of H.M.S. “Swiftshure”

| | |
|---|---|
| **Artist** | C. F. Henderson |
| **Year** | 1884 |
| **Dimensions** | 33 × 20 cm (13 × 7.9 in.) |
| **Location** | David Rumsey Map Collection |

This ship’s log shows how position-finding worked in practice on a Victorian-era ship. The left page records hourly readings of course, speed, wind, and barometric pressure, notes the ship’s latitude and longitude, and includes bearings to landmarks on the Californian coast, such as Cape Mendocino.

The right page shows a hand-drawn coastline sketched from published reference maps and charts, on which the crew plotted the positions recorded in the log. In this way they traced the Swiftshure’s track down the coast from Esquimalt on the southern tip of Vancouver Island to San Francisco Bay.

H.O.A 38
H.O.A. 38
CARY LONDON

# 41. [Cary’s quintant]

| | |
|---|---|
| **Artist** | William Cary |
| **Place** | London |
| **Year** | 1885 |
| **Location** | Paul Saffo Cartographic Instrument Collection |

Quintants were used on board a ship to aid in wide horizontal angles of hydrographic surveying. Similar to a sextant but with a slightly larger arc—one-fifth rather than one-sixth of a circle—a quintant’s reflecting properties allowed it to measure wide angles up to 144°, enabling maritime surveyors to measure between landmarks on the coast and fix a ship’s position at sea. This pocket-sized tool, in a leather case labeled with “H.O.” for the British Hydrographic Office, is the type of instrument which would have been used for the observed locations in the H.M.S. Swiftshure’s log book: taking measurements at sea level where altitude readings are taken from a true horizon and eliminating the need to correct for an observer’s height above the sea.

Camp No. 62 to No. 63 at Lahaina Roads, Topographer [illegible]

Date Nov 26, 1878. Instrument used [illegible]

[illegible] Party.

16°30' Var.

Mag. Var.
N. Lat.
S. Lat.
E. Dep.
W. Dep.
Travel

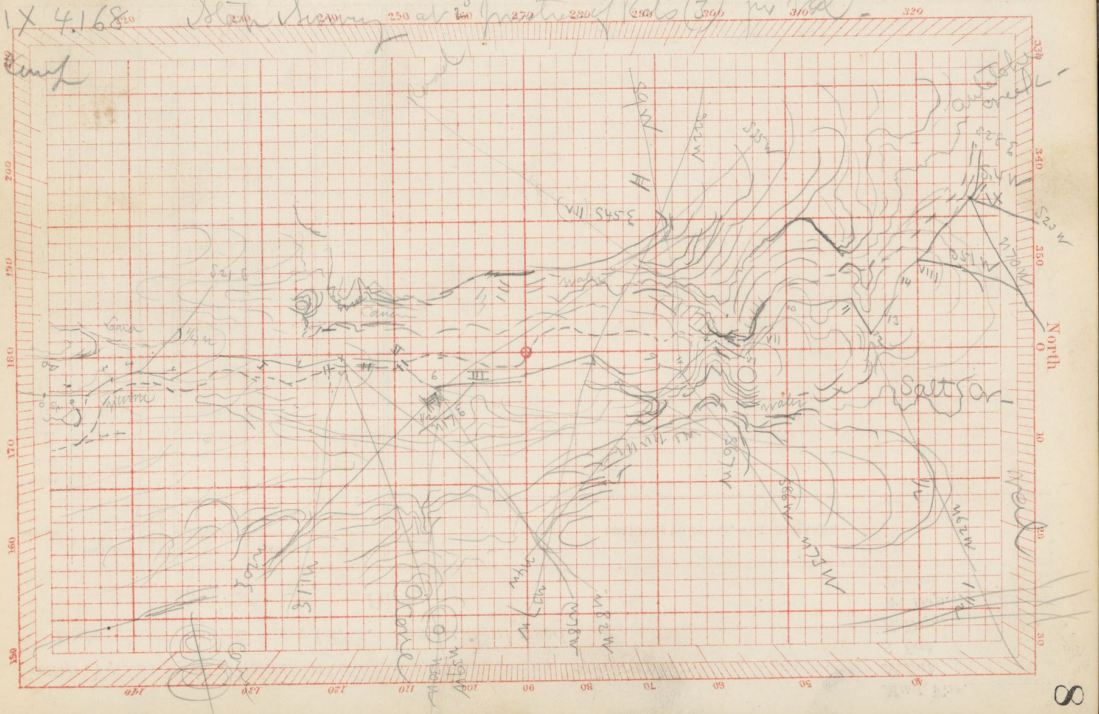
| Station. | Distance by | Per cent. Reduction. | Bearing. | Aneroid No. Inches. | Aneroid No. Feet. | Bearings to Peaks, etc.—Remarks. |
|---|---|---|---|---|---|---|
| Camp. 62 | 73 | | S64W | | | |
| 1 | 170 | -2 | S20W | 17° | | N66W |
| 2 | 330 | -4 | S71W | | | N63W |
| 3 | 622 | | S88W | | | N58½W |
| III | 876 | | S87W | | | N53W |
| 4 | 1064 | | S74W | | | N46W |
| 5 | 1160 | | S71W | | | N39W |
| II | 1270 | | S80W | | | N34W |
| 6 | 1325 | | S80W | | | N30W |
| III | 1793 | | S50W | | | |
| 7 | 1990 | -2 | S74W | | | N9W |
| 8 | 2090 | | S47W | | | |
| 9 | 2122 | | S68W | | | |
| IV | 2269 | | S25W | | | N2E |
| V | 2322 | | S63W | | | |
| VI | 2375 | | S15W | | | N35E |
| VII | 2436 | | S13E | | | N6E |
| 10 | 2557 | -5 | S76W | | | |
| 11 | 2613 | | N80W | | | |
| 12 | 2637 | | S7W | | | |
| 13 | 2703 | | S12E | | | |

1427 19 S3W

VIII 2886 -20 S4W S15E

IX 4168

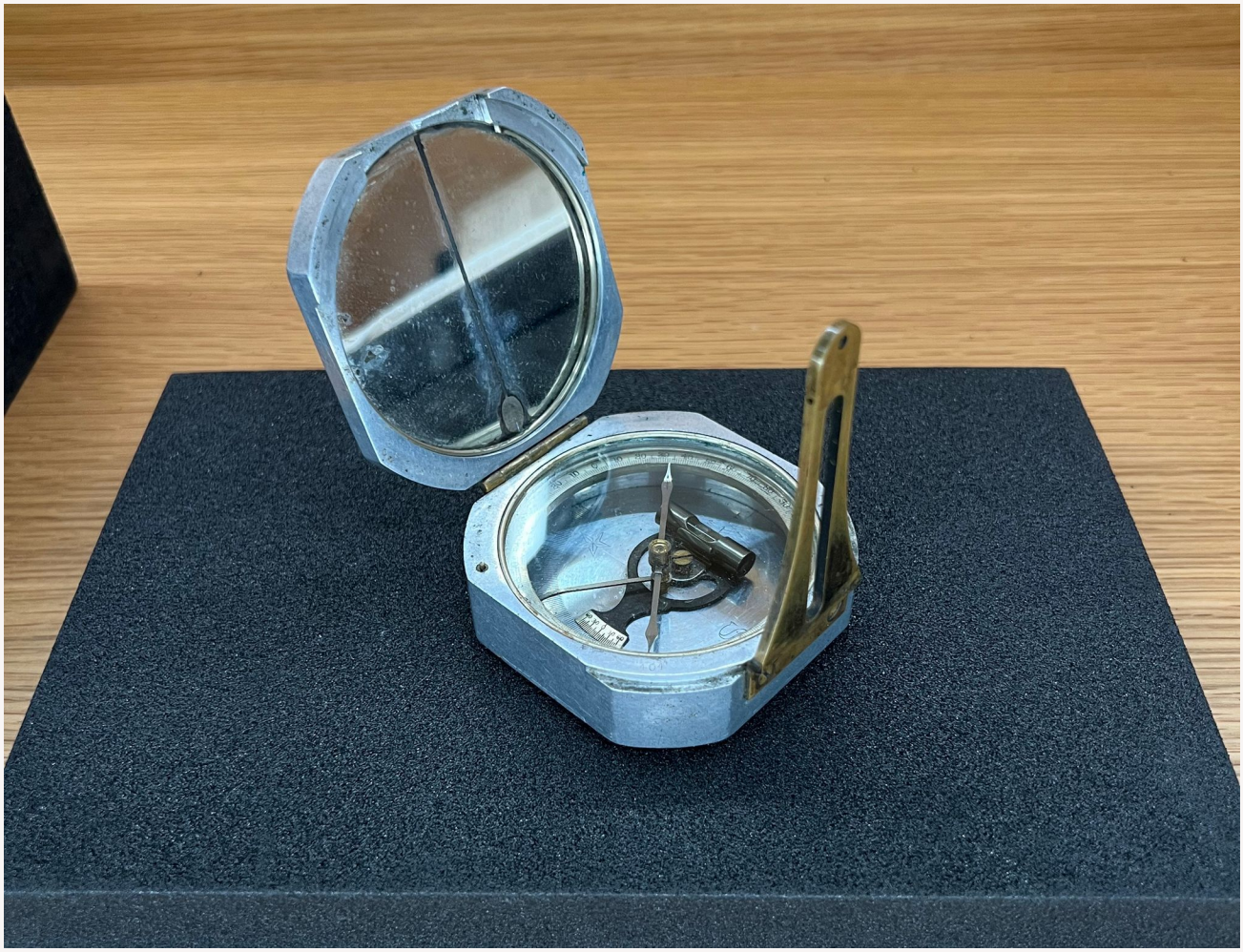
Junction of Roads



# 42. [Manuscript survey journal of Gilbert Thompson, Wheeler Survey]

| | |
|---|---|
| **Artist** | Gilbert Thompson |
| **Year** | 1878 |

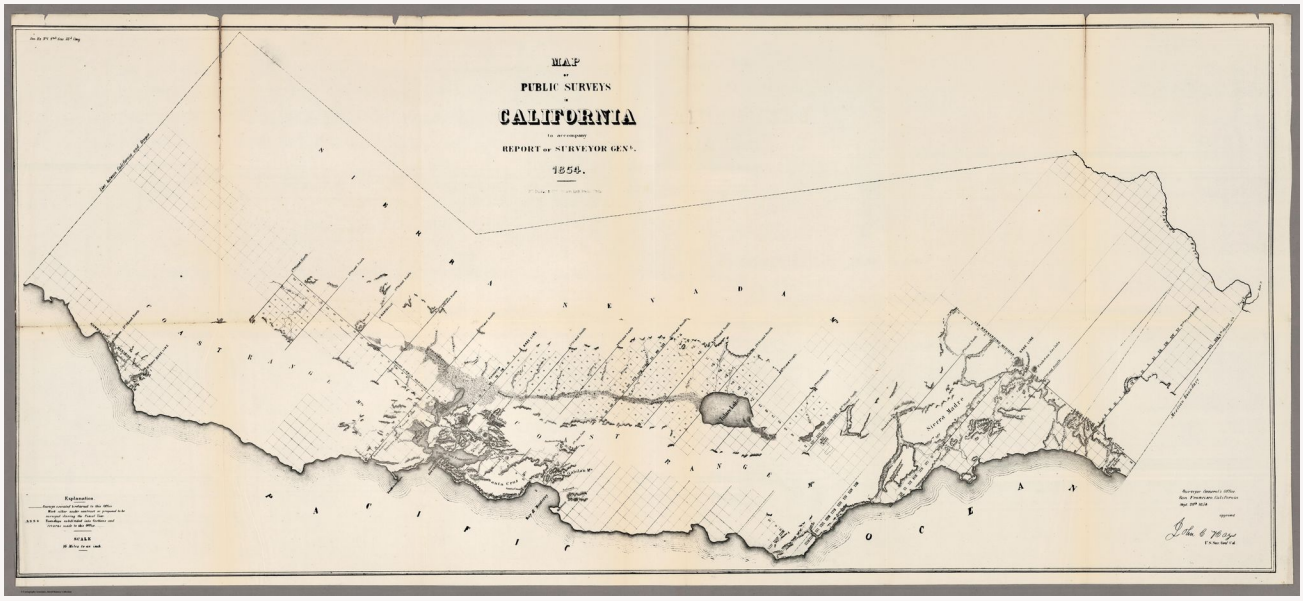
Gilbert Thompson, a topographer with the United States Geographical Surveys West of the 100th Meridian, kept this journal on an 1878 expedition ending in Chico, California. Field notebooks like this one were essential for collecting information for survey maps. On the upper page, key information about each location (or “station”) along the route was captured in a preprinted form: distance traveled, compass bearing, and barometer readings for elevation, along with local magnetic declination (“Mag. Var.”), noted in red, and observations like “arriving at Camp 62, a sheep ranch on the Belle Mill Road”. On the lower page, measurements were combined into a carefully hand-rendered map augmented with a sketch of the profile of the horizon across the top. Records like these allowed a route to be retraced and provided rich details for the cartographers responsible for creating final maps for publication.



# 43. Brunton compass

| | |
|---|---|
| **Artist** | Daniel W. Brunton |
| **Place** | Denver |
| **Year** | 1904 |
| **Location** | Paul Saffo Cartographic Instrument Collection |

David Brunton developed the pocket transit in 1894 as a compact, multi-purpose instrument for field exploration, and partnered with William Ainsworth and Sons soon after to manufacture it. Brunton called it a “transit” and not a standard compass because it does more than point north. This single palm-sized instrument could measure vertical and horizontal angles to a visual target as well as indicate directional bearings, combining a bubble level for aligning the device through gravity with a clinometer showing degrees and percent of slope of the target from the viewer. This specific device is notable because it was made by Brunton in the earliest years of production, as indicated by the serial number 404, retains its working “wet” bubble level, and is known to have once belonged to a Mr. Steed, who worked in the mining industry in Alaska and Argentina.


MAP
of
PUBLIC SURVEYS
in
CALIFORNIA
to accompany
REPORT of SURVEYOR GEN.
1856.
Explanation
SCALE
SIERRA NEVADA
COAST RANGE
PACIFIC OCEAN

# 44. Map of the Public Surveys in California, 1854

| | |
|---|---|
| **Artist** | U.S. General Land Office |
| **Place** | Philadelphia |
| **Year** | 1854 |
| **Dimensions** | 52 x 119 cm (20.5 × 46.9 in.) |
| **Location** | David Rumsey Map Collection |

Land had to be inventoried and measured before it could be sold and settled. After statehood in 1850, surveyors divided California using the Public Land Survey System, a rectangular grid with sections called townships measured almost entirely by hand. This map, published in 1854, was the first of eleven issued annually to show the progress and plans for this system to radiate from three initial points at Mount Diablo, San Bernardino, and Mount Pierce, each a prominent peak that could be referenced by surveyors for miles. Faint lines on this map show where townships were completed, proposed, and further subdivided into smaller sections. Only this first edition is oriented with east at the top, showing California as settlers arriving by sea would have seen it. The grid on this map still reflects roads and property lines today.

# Acknowledgements

| | |
|---|---|
| **Curator** | Evan Thornberry, Head Librarian and Curator |
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| **Digital Exhibition** | Evan Thornberry |
| **Project Manager** | Laura Swenson, Operations Manager |
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